

The Geometry of Nontermination in Randomized Local Search

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Abstract

How much does a randomized algorithm's running-time distribution reveal about its behavior under a different source of randomness? We construct a local search process in which every deterministic rule choosing repairs using only the past has the same stopping-time law under an independent reference source P . The sums of powered prefix probabilities also agree at every depth and every positive exponent. Yet two repair orders have worst-case survival probabilities 1 and $\exp[-\Theta(n)]$ at a deadline of $8n$ steps, over the same class of sources satisfying $\nu[w] \leq P[w]^{1/2}$ for every prefix w . For each n , a single explicit source separates the two orders.

Classical tree max-flow/min-cut identifies the worst failure probability under a global prefix bound with a minimum-cost prefix cover. For disagreement repair on any connected N -vertex graph, we compute this quantity in $O(NT)$ arithmetic operations through horizon T . Starting with one vertex of value one, a fixed four-symbol reference source gives the worst-case survival profile $\exp[-\Theta(\min\{T^{1/3}, N\})]$ under the prefix bound $\nu[w] \leq P[w]^{\Delta_N}$, where Δ_N is the Hausdorff dimension of its nonterminating tapes. The best single-depth bound is only $\Theta(N^{-3})$. We also obtain an exact composition law: $N - 1$ components whose individual worst-case nontermination probabilities are $\exp[-\Theta(N)]$ together admit a single source that forces nontermination under every interleaving, with the same global prefix bound.

The Hausdorff dimension of the nonterminating tapes gives the critical covering exponent. Applying the matrix-commutative local lemma to powered action probabilities yields conditional min-entropy thresholds for k -SAT. A guarded reduction from the tight SAT construction of Gebauer, Szabó, and Tardos shows that the maximum-degree threshold is asymptotically sharp within $o(1)$ bits per repair, even for satisfiable formulas.

1 Introduction

How much can a change in the source of randomness alter a randomized local search procedure? In the procedures studied here, each step selects a violated constraint and attempts to resolve it by changing some variables. We call this update a repair; the selected constraint may remain violated. The deterministic selection rule can use the past but cannot inspect future randomness; we call such a rule nonanticipating. We fix the input and the rule that maps source symbols to updates, and ask which properties of the reference execution determine its behavior under a different source.

Even the complete running-time distribution gives only part of the answer. Consider independent tasks, each retried until a specified symbol completes it. Under our reference source, every repair order has the same stopping-time law. More strongly, the sums of powered prefix probabilities agree at every depth and every positive exponent. Nevertheless, under the same constraint on imperfect randomness, one order can remain unfinished with certainty at a linear deadline while another has exponentially small worst-case survival probability. Thus these sums do not capture all the information needed for robustness.

The missing information lies in the prefix tree of executions: which surviving prefixes extend which others. A prefix cover retains that information by allowing different branches to be cut at different depths. Its cost determines the exact worst-case probability of a tape event under the source constraint. At finite horizons this gives robust running-time bounds; at arbitrarily small scales it gives the Hausdorff dimension of the nonterminating tapes.

1.1 A global constraint on the source

Fix a finite alphabet with at least two symbols. Let P be the law of independent draws from a distribution that gives every symbol positive probability. A tape is an infinite sequence of these symbols. Write $[w]$ for the tapes beginning with a finite word w , and $P[w]$ for their reference probability. We consider the class

$$\mathcal{Q}_{s,D} = \{\nu : \nu \text{ is a probability law and } \nu[w] \leq DP[w]^s \text{ for every finite } w\}, \quad 0 < s \leq 1, \quad D \geq 1. \quad (1)$$

The same constant D applies at every prefix length. Smaller s allows a source to concentrate more probability on individual words.

For example, if one source symbol is a block of k fair bits, then a word of T blocks has probability 2^{-kT} , and the constraint is $\nu[w] \leq D2^{-ksT}$. A conditional bound 2^{-ks} on the probability of every next block implies this constraint with $D = 1$; equivalently, each block has conditional min-entropy at least ks bits. The global prefix constraint is weaker. It allows dependence across blocks and does not impose a conditional entropy lower bound at every step.

Let $\tau(\omega)$ be the number of repairs on tape ω , and let $\mathcal{N} = \{\omega : \tau(\omega) = \infty\}$. Let $\mathcal{L}_T$ be the prefix-free family of minimal source prefixes completing T repairs, including a possible final repair that terminates the process. Define the power sum

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s, \quad s > 0. \quad (2)$$

These cylinders are disjoint, so

$$Z_T(1) = P\{\tau \geq T\}, \quad \mathbb{P}_\nu\{\tau \geq T\} \leq DZ_T(s).$$

The second inequality needs domination only on the prefixes in $\mathcal{L}_T$. Exact robustness requires the bound on every prefix, as in $\mathcal{Q}_{s,D}$.

If the reference source is uniform on q symbols and one symbol is read per repair from a fixed initial state, then

$$Z_T(s) = |\mathcal{L}_T| q^{-sT} = q^{(1-s)T} P\{\tau \geq T\}. \quad (3)$$

Thus the ordinary stopping-time law determines all these sums for a uniform reference source. That implication need not hold for a nonuniform source. Our nonuniform example nevertheless preserves every power sum.

1.2 Exact robustness through prefix covers

The bound above uses prefixes that all reach the same repair count. To cover a tape event E , one can instead choose prefixes of different lengths. For a closed set E and $0 < s \leq 1$, define

$$\mathcal{C}_s(E) = \inf_{\mathcal{B}} \sum_{w \in \mathcal{B}} P[w]^s, \quad E \subseteq \bigcup_{w \in \mathcal{B}} [w], \quad (4)$$

where $\mathcal{B}$ ranges over finite prefix-free covers. The empty prefix has cost one. Classical tree max-flow/min-cut [BP17, Section 3.1] gives the following interpretation.

Theorem 1.1 (Exact source bound from prefix covers). *For every closed E , $0 < s \leq 1$, and $1 \leq D < \infty$,*

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu(E) = \min\{1, D\mathcal{C}_s(E)\}. \quad (5)$$

If E depends on the first T symbols, the infimum is attained by a cover using prefixes of length at most T . If finite-symbol events E_T decrease to $\mathcal{N}$, then

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu(\mathcal{N}) = \lim_{T \rightarrow \infty} \max_{\nu \in \mathcal{Q}_{s,D}} \nu(E_T). \quad (6)$$

A source sends probability flow down the prefix tree, and a cover cuts every branch in E . Appendix A proves the identity, including completion of a maximizing subflow to a probability law. Apply the identity to survival beyond a deadline or to $E = \mathcal{N}$.

For distinct tapes, put $d_P(\omega, \eta) = P[\omega \wedge \eta]$, where $\omega \wedge \eta$ is their longest common prefix, and set $d_P(\omega, \omega) = 0$. Each cylinder $[w]$ then has diameter $P[w]$. The Hausdorff dimension $\delta = \dim_H^P \mathcal{N}$ is the critical exponent for covering $\mathcal{N}$ at small scales [Bil65, Lut11]. For $s < \delta$, some finite D permits a source in $\mathcal{Q}_{s,D}$ supported on $\mathcal{N}$. For $s > \delta$, every source in this class gives $\mathcal{N}$ probability zero. At equality, the cover content matters: the smallest domination constant permitting certain nontermination is $1/\mathcal{C}_s(\mathcal{N})$, with $1/0 = \infty$. Dimension identifies an asymptotic threshold; quantitative covers are needed to determine finite-time behavior.

1.3 Contributions and proof ideas

Repair order. Section 2 proves the separation for independent retry tasks. Normalizing powered symbol probabilities turns the total work into a sum of independent geometric workloads, proving equality of all power sums. A cover at different depths and an explicit admissible source prove the robustness separation. Optimizing the cover recursion computes an optimal deterministic selector in $O(nT)$ arithmetic operations for this example.

Exact finite-time bounds and composition. Section 3 studies updates of graph edges whose endpoints disagree. The number of ones determines the stopping time on each tape and reduces the exact cover calculation to $O(NT)$ operations. For a fixed four-symbol reference source, starting with one vertex of value one, the worst survival probability under $\mathcal{Q}_{\Delta_N,1}$ is $\exp[-\Theta(\min\{T^{1/3}, N\})]$, where Δ_N is the critical exponent. The proof balances the cost of reaching an intermediate weight against the cost of staying below it. This gives the sharp N^3 time scale, while every single-depth bound is at least of order N^{-3} . The same harmonic potential gives an exact composition law: one source can make $N - 1$ components, each with worst-case nontermination probability $\exp[-\Theta(N)]$ in isolation, run forever under every interleaving. A further construction preserves ordinary transition matrices while changing the nontermination dimension from near zero to near one.

Entropy thresholds for satisfiability search. Section 4 proves exponential termination when each redraw has conditional min-entropy greater than $\log_2 \kappa_D$, where κ_D is the trace-growth rate of the clause-overlap graph. At maximum degree $d \geq 2$, more than $\log_2 c_d$ bits per redraw suffice, where $c_d = d^d / (d - 1)^{d-1}$. Adding a private guard to each clause of the unsatisfiable formulas of Gebauer, Szabó, and Tardos [GST16] makes them satisfiable while preserving a nonterminating source within $o(1)$ bits of this threshold.

Underlying methods. Tree capacity and the links between weighted growth and dimension are classical [BP17, Sta93, MW88]. The local upper bounds apply the matrix method of Harris, Iliopoulos, and Kolmogorov [HIK25] to powered action probabilities. Cuts at different trajectory depths also appear in entropy compression [AIV17]. Section 5 gives the detailed comparisons.¹

2 Scheduling under correlated randomness

Consider n unfinished tasks of type A and one of type B . A repair chooses an unfinished task and reads one binary symbol. An A task completes on 0; the B task completes on 1. Otherwise the task remains unfinished. Completed tasks are never selected again. Updating one task does not change another task. Selection may use the entire observed history.

Theorem 2.1 (Equal prefix sums, different robustness). *Fix $P(0) = 15/16$, $P(1) = 1/16$, and the common class $\mathcal{Q} = \mathcal{Q}_{1/2,1}$. Let Λ_A complete all A tasks before attempting B , and let Λ_B complete B first. For every $n \geq 1$,*

$$\max_{\nu \in \mathcal{Q}} \mathbb{P}_{\nu}^{\Lambda_A} \{\tau > 8n\} = 1, \quad (7)$$

$$\left(\frac{15}{16}\right)^{4n} \leq \max_{\nu \in \mathcal{Q}} \mathbb{P}_{\nu}^{\Lambda_B} \{\tau > 8n\} \leq \frac{9}{8} \left(\frac{225}{256}\right)^n < 1. \quad (8)$$

One tape law $\nu_n \in \mathcal{Q}$ gives survival probability one for Λ_A and exponentially small survival probability for Λ_B . Nevertheless, every deterministic nonanticipating selector has the same $Z_T(s)$ for every $s > 0$ and every $T \geq 0$, and the same surviving-prefix sum $\tilde{Z}_T(s)$ for every s, T . Here $\tilde{Z}_T$ sums powered probabilities of prefixes still live after T repairs, whereas Z_T includes a repair that terminates at time T . In particular, every selector has the same complete stopping-time law under P . The lower bound in (8) holds for every selector. Thus the optimal worst-case survival probability is $\exp[-\Theta(n)]$, and Λ_B attains this exponential order.

¹ChatGPT and Codex assisted with developing and checking the scheduling, prefix-cover, disagreement, composition, and guarded-SAT results, and with literature comparisons and exposition. The AI methodologies statement at the end describes their contributions.

The full collection of single-depth power sums therefore cannot determine the worst survival probability under a global prefix constraint. The missing information is how the surviving prefixes nest. In this example, changing the repair order exploits that information at a linear deadline. For every fixed n and every deterministic selector, the nonterminating tapes form a countable set and have Hausdorff dimension zero. Every source in $\mathcal{Q}$ gives this set probability zero. The separation therefore concerns finite time, which dimension alone does not determine.

Why all prefix power sums agree. For $s > 0$, normalize $P(0)^s, P(1)^s$ to probabilities and call the resulting i.i.d. law P_s . With $W_s = P(0)^s + P(1)^s$,

$$\tilde{Z}_T^\Lambda(s) = W_s^T \mathbb{P}_{P_s}^\Lambda \{\tau > T\}.$$

Preassign an independent stream to each task and reveal its next symbol when it is selected. The total number of attempts is a sum of independent geometric workloads, regardless of their order. This proves equality under every P_s , including selectors that remember past symbols.

Why the robust tails differ. To defeat Λ_A , generate 0 with probability $3/4$ until the n th zero, then force zeros through time $8n$. Continue with P afterwards. This uses randomness early to make a later deterministic stretch admissible under the global constraint. The B task then never sees its successful symbol before the deadline.

For Λ_B , cut first at 0^{4n} , whose power cost is $(225/256)^n$. If the first 1 occurs by time $4n$, then B completes; survival requires fewer than n zeros in the next $4n$ symbols. A generating-function bound makes the total cost of these remaining cylinders at most $\frac{4+\sqrt{15}}{3\sqrt{15}}(\sqrt{15}/27)^n$. This is a cover at different depths. It uses the bound on whole prefixes, without assuming a conditional bound after the first 1. Appendix B.1 checks every prefix of the constructed source and proves both estimates.

Computing a robust repair order. For $0 < s \leq 1$, take a finite state space, a common finite action alphabet with full-support i.i.d. probabilities $\pi(a)$, and a finite nonempty set of available repairs F_x at each live state. Write $P = \pi^\mathbb{N}$ and let $\Phi_f(x, a)$ be the state after repair f uses symbol a at state x . Define $V_0(x) = 1$ at live states and $V_t(x) = 0$ at terminated states. At live states set

$$V_{t+1}(x) = \min \left\{ 1, \min_{f \in F_x} \sum_a \pi(a)^s V_t(\Phi_f(x, a)) \right\}. \quad (9)$$

Proposition 2.2 (Optimal deterministic selection). *For every $D \geq 1$ and finite horizon T ,*

$$\min_{\Lambda} \max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_{\nu}^\Lambda \{\tau > T\} = \min\{1, DV_T(x_0)\}, \quad (10)$$

where Λ ranges over deterministic nonanticipating selectors. One selector depending only on current state and remaining time is optimal simultaneously for every $D \geq 1$ and every initial state.

The proof minimizes the finite-tree cover recursion; Appendix B.1 gives the induction. For the task example, only the number of unfinished A tasks and whether B remains are needed, so the optimum uses $O(nT)$ arithmetic operations. We do not need to identify an exact optimal order to prove Theorem 2.1.

3 Exact bounds for a disagreement process

We next consider a process in which repairs overlap. Vertices carry zero or one, and the algorithm updates edges whose endpoints disagree. A single count determines its stopping time on every tape. This gives exact dimensions and worst-case probabilities under the prefix constraint.

Let $G = (V, E)$ be a connected graph with $N = |V| \geq 2$, and give every edge a fixed orientation. A state is $x \in \{0, 1\}^V$, and an edge is flawed when its endpoints disagree. Let $\mathcal{A}$ be a finite action alphabet with a full-support law π , and write $P = \pi^{\mathbb{N}}$ for the action-tape law (the initial assignment is fixed). Partition the alphabet into four nonempty classes $\mathcal{A}_-, \mathcal{A}_p, \mathcal{A}_r, \mathcal{A}_+$. When a selected oriented edge reads $z \in \{01, 10\}$, a symbol in these four classes replaces z , respectively, by 00, by z , by the reversed word $\bar{z}$, or by 11. Thus several action labels may produce the same state transition. The tape consists of independent draws from π , and any deterministic nonanticipating rule may choose the next flawed edge, even using the earlier action labels.

For $s > 0$, define the three powered action weights

$$D_s = \sum_{a \in \mathcal{A}_-} \pi(a)^s, \quad N_s = \sum_{a \in \mathcal{A}_p \cup \mathcal{A}_r} \pi(a)^s, \quad U_s = \sum_{a \in \mathcal{A}_+} \pi(a)^s. \quad (11)$$

This is a four-outcome extension of discordant voting [CDFR18]. Conditional on a selected discordant edge, the oblivious voting update is the boundary case with mass $1/2$ on each of 00 and 11: the selected edge becomes monochromatic. Here the selected edge may remain discordant, either unchanged or reversed, and the edge selector may be any deterministic nonanticipating rule. Cooper, Dyer, Frieze, and Rivera [CDFR18] study expected consensus time; the result below determines the complete stopping-time law and nontermination dimension for the extended rule.

3.1 The exact calculation

For the four-vertex path P_4 , Hamming weight reduces the fourteen live assignments to three states:

$$\begin{aligned} P_4 : \quad & \{x \in \{0, 1\}^4 : x \text{ is live}\} \xrightarrow{M(x) = \sum_v x_v} \{1, 2, 3\}, \\ 0_{\text{stop}} & \xleftarrow{D_s} 1 \xleftarrow{\frac{U_s}{D_s}} 2 \xleftarrow{\frac{U_s}{D_s}} 3 \xrightarrow{U_s} 4_{\text{stop}}, \quad R_s(k, k) = N_s. \end{aligned} \quad (12)$$

Theorem 3.1 (Stopping times and dimension). *Start from a fixed live assignment with k_0 ones. For $s > 0$, let R_s be the $(N-1) \times (N-1)$ tridiagonal matrix*

$$R_s(k, k-1) = D_s, \quad R_s(k, k) = N_s, \quad R_s(k, k+1) = U_s, \quad (13)$$

where entries outside $\{1, \dots, N-1\}$ are omitted, and put $S_s = D_s + N_s + U_s = \sum_{a \in \mathcal{A}} \pi(a)^s$. For every deterministic nonanticipating selector,

$$Z_0(s) = 1, \quad Z_T(s) = S_s e_{k_0}^T R_s^{T-1} \mathbf{1} \quad (T \geq 1), \quad (14)$$

For fixed N , k_0 , and action partition, the stopping time agrees on every tape, independently of G and of the selector. With $c_N = \cos(\pi/N)$, the exact exponential growth rate is

$$\varrho_s = \rho(R_s) = N_s + 2(D_s U_s)^{1/2} c_N. \quad (15)$$

Under the reference source P , the process terminates almost surely with an exponential tail, and

$$\dim_H^P \mathcal{N} = \Delta, \quad N_\Delta + 2(D_\Delta U_\Delta)^{1/2} c_N = 1, \quad (16)$$

where the equation has a unique solution $\Delta \in (0, 1)$. If the probabilities $\pi(a)$ are computable, the effective strong dimension defined in (39) satisfies

$$\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega) = \Delta, \quad (17)$$

and the supremum is attained, even if the selector is not computable.

The same Δ is the exact threshold for global prefix domination. If a possibly dependent action source ν satisfies

$$\nu[w] \leq DP[w]^s \quad (w \in \mathcal{L}_T, T \geq 0) \quad (18)$$

for some $D < \infty$ and $s > \Delta$, then

$$\mathbb{P}_\nu\{\tau \geq T\} \leq DS_s e_{k_0}^\top R_s^{T-1} \mathbf{1} = O(\varrho_s^T) \quad (T \geq 1). \quad (19)$$

At $s = \Delta$, there is a law ν_Δ on the same action tree, supported on $\mathcal{N}$, and a constant $B_\Delta < \infty$ for which $\nu_\Delta[w] \leq B_\Delta P[w]^\Delta$ for every prefix w ; since $P[w] \leq 1$, the same law is dominated at every $0 < s \leq \Delta$.

Proof sketch. Every flawed edge contains exactly one vertex of value one. An action therefore changes Hamming weight by -1 , 0 , or $+1$, with total powered probability mass D_s , N_s , or U_s , independently of the selected edge. The sum over prefixes still live after T repairs is $e_{k_0}^\top R_s^T \mathbf{1}$. To complete $T \geq 1$ repairs, the process must remain live after $T-1$ repairs and may then use any action; its total last-step mass is S_s . This proves (14). A diagonal similarity turns R_s into a symmetric tridiagonal matrix with off-diagonal entry $(D_s U_s)^{1/2}$, yielding (15). The finite-state pressure formula and its Perron–Doob law then give the two dimensions and the sharp source-domination threshold.

The proof, including the critical law attaining the threshold and both dimensions, is in Appendix B.2.

3.2 Exact robustness at finite horizons

The same Hamming-weight reduction computes the optimal prefix cover. For $0 < s \leq 1$, define vectors indexed by $1, \dots, N-1$ through

$$C_{0,s} = \mathbf{1}, \quad C_{T+1,s} = \min\{\mathbf{1}, R_s C_{T,s}\}, \quad (20)$$

where the minimum is coordinatewise. At each prefix, the recursion chooses between covering its whole cylinder and covering its surviving children.

Theorem 3.2 (Exact robustness in $O(NT)$ operations). *Start from a fixed live assignment with k_0 ones. For every connected N -vertex graph, every deterministic nonanticipating selector, $0 < s \leq 1$, and $D \geq 1$,*

$$\max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_\nu\{\tau > T\} = \min\{1, DC_{T,s}(k_0)\}. \quad (21)$$

After computing D_s, N_s, U_s , all values through horizon T require $O(NT)$ arithmetic operations and $O(N)$ working storage. For completion of $T \geq 1$ repairs, use $C_{T-1,s}$.

At the critical power Δ of Theorem 3.1, put $h(k) = (D_\Delta/U_\Delta)^{k/2} \sin(k\pi/N)$ and $h_{\max} = \max_{1 \leq j < N} h(j)$. Then

$$C_{T,\Delta}(k) \downarrow \frac{h(k)}{h_{\max}}, \quad \max_{\nu \in \mathcal{Q}_{\Delta,D}} \nu(\mathcal{N}) = \min\left\{1, D \frac{h(k_0)}{h_{\max}}\right\}. \quad (22)$$

Thus the smallest domination constant permitting certain nontermination is exactly $h_{\max}/h(k_0)$.

The recursion is exact even when the selector remembers all past action labels: their future weight increments depend only on the current weight. At the critical power, a positive left Perron vector forces the limiting fixed point to be harmonic, yielding the displayed ratio. The full proof is in Appendix B.3.

Theorem 3.3 (Matching finite-time robustness bounds). *Use four action labels with the fixed probabilities*

$$\pi_- = \frac{49}{100}, \quad \pi_p = \pi_r = \frac{1}{4}, \quad \pi_+ = \frac{1}{100}. \quad (23)$$

For $N \geq 3$, start disagreement repair on any connected N -vertex graph at Hamming weight one, and let Δ_N be its critical power. Then $1/2 < \Delta_N < 1$. There are absolute constants $0 < c < C < \infty$ such that, for every integer $T \geq 1$ and every deterministic nonanticipating selector,

$$e^{-Cq} \leq \max_{\nu \in \mathcal{Q}_{\Delta_N, 1}} \mathbb{P}_\nu\{\tau > T\} = C_{T, \Delta_N}(1) \leq e^{-cq}, \quad q = \min\{T^{1/3}, N\}. \quad (24)$$

Its limit is exactly

$$\max_{\nu \in \mathcal{Q}_{\Delta_N, 1}} \nu(\mathcal{N}) = 7^{-\Delta_N(N-2)}. \quad (25)$$

In contrast, every single-depth surviving-prefix sum satisfies

$$\tilde{Z}_t(\Delta_N) := e_1^\top R_{\Delta_N}^t \mathbf{1} \geq \frac{2}{N} \sin^2 \frac{\pi}{N} \quad (t \geq 0), \quad (26)$$

and its infimum over t is $\Theta(N^{-3})$. All constants are uniform over N , graphs, and selectors.

Thus $T = \Theta(N^3)$ is the necessary and sufficient scale for worst-case survival $\exp[-\Theta(N)]$. The same four-symbol reference source works for every N ; its critical exponent Δ_N depends on N . The maximizing alternative source may depend on the horizon. Paths P_N already realize the exponential gap with overlapping repairs. Since $Z_T(\Delta_N) = S_{\Delta_N} \tilde{Z}_{T-1}(\Delta_N)$ for $T \geq 1$, the completed-repair sums also give only a polynomial bound at their best depth.

Why the cube root appears. For $2 \leq m \leq N/2$, cut the prefix tree when the weight first reaches m , or at time T if it remains below m . The harmonic cost of the first part is $\exp[-\Omega(m)]$; the spectral cost of the second is $\exp[-\Omega(T/m^2)]$. Balancing them gives $m \asymp T^{1/3}$. A Perron vector on weights $1, \dots, m-1$ gives the matching lower bound $\exp[-O(m + T/m^2)]$. Once m reaches the full interval, the critical harmonic vector supplies the limiting value. Appendix B.4 gives the uniform bounds, including small N and T , and a refinement reaching twice the exact limiting probability by $3N^4$ repairs.

3.3 Composing repair components

The critical calculation extends to arbitrarily interleaved components. Count the ones in components that have not yet become constant. An upward repair that makes a component all ones removes that entire component from the count. This can cause a large downward jump, so the count no longer determines the entire transition law.

Theorem 3.4 (Exact robustness of many repair components). *Fix $N \geq 3$ and use the source $(49/100, 1/4, 1/4, 1/100)$ and exponent Δ_N from Theorem 3.3. Let G be a disjoint union of finitely many connected graphs, each with at least N vertices. Run disagreement repair, choosing any currently disagreeing edge by an arbitrary deterministic nonanticipating selector. In particular, the selector may interleave the components and remember all past labels. Let K_0 be the number of ones in the initially nonconstant components; constant components are ignored. Put*

$$\theta = \pi/N, \quad r = 7^{\Delta_N}, \quad V_N(k) = \begin{cases} 0, & k = 0, \\ r^{k-N+1} \frac{\sin(k\theta)}{\sin \theta}, & 1 \leq k < N-1, \\ 1, & k \geq N-1. \end{cases}$$

Then, for every $D \geq 1$,

$$\max_{\nu \in \mathcal{Q}_{\Delta_N, D}} \mathbb{P}_\nu\{\tau = \infty\} = \min\{1, DV_N(K_0)\}. \quad (27)$$

For each N, K_0, D , one source depending only on these parameters attains this value simultaneously for every allowed graph, initial assignment with count K_0 , and selector. The capacity $V_N(K_0)$ is computable from N and K_0 , although the unresolved-one count does not determine the transition law of the process. Under the reference i.i.d. source,

$$\mathbb{E}_P \tau \leq \frac{25}{12} K_0. \quad (28)$$

For M components each starting with one vertex equal to one, $K_0 = M$. In isolation each component has worst nontermination probability $r^{-(N-2)} = \exp[-\Theta(N)]$ under $\mathcal{Q}_{\Delta_N, 1}$. Together, one source makes $N-1$ components run forever under every interleaving, with that same global source constraint, although $\mathbb{E}_P \tau \leq 25M/12$. The upper bound cuts at $K = N-1$; below this level, no component can complete upward. The matching source forbids upward moves at and above that level and uses harmonic transition probabilities below it. Thus the count follows the same action-driven path under every interleaving. Appendix B.5 gives the construction without restarting a conditional source bound at component boundaries.

3.4 Two implementations of the same transition rule

Corollary 3.5 (The same transition matrices, different dimensions). *For every $\varepsilon > 0$, there exist a computable finite source P and two disagreement-repair maps H and L on P_4 with the following properties. Their ordinary repair matrices (the probabilities of the next state when a specified flaw is repaired) agree for every flaw and state, $K_{f,1}^H = K_{f,1}^L$. Regardless of which deterministic nonanticipating selector is used in either process, including one that remembers the action labels, the stopping law depends only on the initial Hamming weight and is common to H and L . Nevertheless, for every common live initial assignment and every such pair of selectors, their sets of nonterminating tapes, measured in the same source metric, satisfy*

$$\dim_H^P \mathcal{N}_L = \Delta_L < \varepsilon, \quad \dim_H^P \mathcal{N}_H = \Delta_H > 1 - \varepsilon. \quad (29)$$

The values Δ_L and Δ_H are also the attained maximal effective strong dimensions for every such pair of selectors, as defined in (39).

Moreover, the choices can be made so that a single independent identically distributed tape source ν_0 satisfies

$$\nu_0[w] \leq P[w]^{s_0} \quad \text{for every finite action word } w, \quad \max\{\Delta_L, 1 - \varepsilon\} < s_0 < \Delta_H.$$

Under this same source, H never terminates, whereas L stops after exactly $k_0 \leq 3$ repairs, where k_0 is the initial Hamming weight. One source ν_0 works for every live initial assignment and every pair of deterministic nonanticipating selectors.

The construction moves an m -way collision of action labels between a neutral and a downward transition, while preserving every ordinary transition probability. The common alternative source uses only those m labels. It therefore leaves one implementation unchanged and reduces the other's Hamming weight at each repair. Appendix B.6 gives the construction and the dimension calculation. This separates ordinary transition data from source powers; Theorem 2.1 separates all single-depth prefix power sums from prefix-cover robustness.

4 Entropy thresholds for satisfiability search

Throughout, a k -CNF has exactly one literal on each of k distinct variables in every clause. A repair selects a violated clause and redraws its k variables. The selector may use the entire past but cannot inspect future randomness. Let D join two clauses when they share a variable. A repair trace is a sequence of clause names considered up to swaps of adjacent clauses with disjoint scopes; it records the chosen clauses, not the source symbols. Let $\mathcal{M}(D)$ be the set of traces. We assume the formula has at least one clause; an empty formula terminates immediately. Write

$$P_D(\mathbf{z}) = \sum_{I \text{ independent in } D} (-1)^{|I|} \prod_{A \in I} z_A.$$

The classical trace identity says that the generating function of trace counts is $1/P_D(z\mathbf{1})$ [CF69].

For a nonempty finite graph D , let

$$\kappa_D = \limsup_{T \rightarrow \infty} |\{h \in \mathcal{M}(D) : |h| = T\}|^{1/T}. \quad (30)$$

Equivalently, $\kappa_D = R_D^{-1}$, where R_D is the first positive zero of $z \mapsto P_D(z\mathbf{1})$.

Here conditional min-entropy means a pointwise bound: at every feasible positive-probability history $\mathcal{H}_{t-1}$, the next redraw B_t satisfies

$$\max_b \mathbb{P}\{B_t = b \mid \mathcal{H}_{t-1}\} \leq 2^{-h}.$$

Theorem 4.1 (Graph-specific min-entropy threshold for k -SAT). *Let Φ be a CNF on n variables whose clauses each contain exactly k distinct variables, with one literal per variable, and let D be its clause-overlap graph. Start from a fixed assignment, and use any deterministic nonanticipating selector. Let P be the law of independent uniform k -bit redraws. If a possibly history-dependent redraw law ν satisfies, for some $D_0 < \infty$ and $s > 0$,*

$$\nu[w] \leq D_0 P[w]^s \quad (31)$$

for every $w \in \mathcal{L}_T$ and $T \geq 0$, then, for every $\xi > \kappa_D$,

$$\mathbb{P}_\nu\{\tau \geq T\} \leq \frac{D_0 2^n}{P_D(\xi^{-1}\mathbf{1})} (\xi 2^{-ks})^T. \quad (32)$$

In particular, the tail is exponential whenever

$$s > \frac{\log_2 \kappa_D}{k}. \quad (33)$$

If $h > 0$ and each redraw has conditional min-entropy at least h , then (32) holds with $D_0 = 1$ and $ks = h$. Consequently $h > \log_2 \kappa_D$ implies exponential termination. For the uniform reference law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 \kappa_D}{k} \right\}. \quad (34)$$

If the selector is computable, the same bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$.

If D has maximum degree at most $d \geq 2$, then

$$\kappa_D \leq c_d, \quad c_d = \frac{d^d}{(d-1)^{d-1}}. \quad (35)$$

Thus $h > \log_2 c_d$ is a selector-uniform sufficient condition, and the right side of (34) is at most $\min\{1, \log_2(c_d)/k\}$.

In particular, any integer $r > \log_2 c_d$ coordinates of the selected clause may be chosen using the past and redrawn with fresh fair bits, leaving the others unchanged, provided $r \leq k$. The resulting redraw has 2^r equiprobable outcomes at each step, so the theorem applies.

The one-step hypothesis implies global cylinder domination by the chain rule. The converse need not hold: a global bound can redistribute randomness across times. We give separate sharpness results for these two source conditions.

4.1 A sharp obstruction from one fixed bit

The degree-only one-step bound is asymptotically sharp as well. The construction below adds one private variable to each clause of the unsatisfiable formulas of Gebauer, Szabó, and Tardos [GST16, Theorem 1.2]. Fixing that one redraw bit at zero prevents termination, even though all the remaining bits are independent and fair.

Theorem 4.2 (Asymptotic sharpness with one fixed redraw bit). *For every sufficiently large integer r , there is a satisfiable $(r+1)$ -CNF Ψ_r with M_r clauses and maximum clause-overlap degree*

$$d_r = \left(\frac{1}{e} + O(r^{-1/2}) \right) 2^r. \quad (36)$$

Each clause has a distinct private variable. Start with all private variables zero and any assignment of the other variables. For every deterministic nonanticipating selector, the following hold.

If each repair redraws the other r variables with fresh independent fair bits and sets the private variable to zero, every run is infinite. The redraw blocks are independent and have conditional min-entropy exactly r . In contrast, under independent uniform $(r+1)$ -bit redraws,

$$\mathbb{E}\tau \leq 2M_r, \quad \mathbb{P}\{\tau > T\} \leq \mathbb{P}\{\text{Bin}(T, 1/2) < M_r\}, \quad T = 0, 1, \dots \quad (37)$$

In particular, the latter tail is at most $\exp(-T/16)$ for $T \geq 4M_r$. Moreover,

$$0 \leq \log_2 c_{d_r} - r = O(r^{-1/2}) = o(1). \quad (38)$$

Thus no fixed positive number of bits can be subtracted from the degree-only sufficient threshold $\log_2 c_d$ uniformly over finite instances.

This is asymptotic sharpness along the sequence d_r , rather than an exact threshold for every fixed degree. The unsatisfiable formulas and their degree estimate come from Gebauer, Szabó, and Tardos [GST16]; the additional step is the conversion to a satisfiable instance with a nonterminating redraw source.

Construction. Add a private positive literal to every clause of the unsatisfiable formula. This makes the formula satisfiable and preserves its overlap graph. Keeping each private bit zero prevents termination. Under fair redraws, a private bit becomes one with probability $1/2$ and permanently satisfies its clause, so M_r such successes suffice. The full proof and the entropy comparison are in Appendix D.1.

4.2 The dimension of a run and of the exceptional set

For a computable full-support finite source, the effective strong dimension is

$$\text{Dim}_{\text{eff},P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{-\log_2 P[\omega_{1:n}]}, \quad (39)$$

where K is prefix-free Kolmogorov complexity [Sta93, AHLM07, Lut11]. It measures the fraction of source information retained by a tape. It is distinct from the local dimension of a measure, which requires a specified measure on the exceptional set.

For an infinite run ω , let

$$\text{Rec}(\omega) = \{A : A \text{ is repaired infinitely often on } \omega\}. \quad (40)$$

This set is nonempty because there are finitely many clauses.

Corollary 4.3 (Pointwise localization to recurrent clauses). *For a computable repair process with exact clause width k , as in Theorem 4.1, and every infinite run ω ,*

$$\text{Dim}_{\text{eff},P}(\omega) \leq \min \left\{ 1, \frac{\log_2 \kappa_{D[\text{Rec}(\omega)]}}{k} \right\}. \quad (41)$$

For each nonempty set R of clauses, the same right side with $D[R]$ bounds the Hausdorff dimension of the tapes satisfying $\text{Rec}(\omega) = R$.

Clauses repaired only finitely often affect a finite prefix but not the asymptotic information bound. The proof fixes that prefix and applies the graph-specific bound to the remaining clauses. No assumption that the recurrent set can be computed from a finite execution is needed.

Sharp dimension bounds on trees. For $2 \leq d \leq k$, Theorem D.6 constructs satisfiable k -CNFs whose overlap graphs are finite rooted $(d-1)$ -ary trees $T_{d,\ell}$. With a uniform initial assignment and a current-assignment selector, they satisfy

$$\dim_H^P \mathcal{N} = \frac{\log_2 \kappa_{T_{d,\ell}}}{k}, \quad \kappa_{T_{d,\ell}} \uparrow c_d \quad (\ell \rightarrow \infty).$$

The same exponent is the maximal effective strong dimension for a computable selector. A source supported on $\mathcal{N}$ attains global cylinder domination at that exponent. Thus the degree-based dimension bound is sharp in the supremum over finite instances. The full statement and construction appear in Appendix D.2.

The tree formulas are extremal: overlapping bad events cannot occur together. Theorem D.6 uses their established trace structure to attain the dimension and global source bounds. A clique construction in Proposition D.4 attains the one-step threshold $\log_2(d+1)$ for its particular graph. The guarded construction above supplies the different assertion of asymptotic sharpness for the degree-only one-step threshold.

Further consequences, including explicit tail constants, adaptive redraws using $O(\log d)$ fresh bits, Santha–Vazirani sources, and a matching repair oracle, are in Appendix D.

5 Related work

Kolmogorov [Kol18] developed action-level commutativity for resampling oracles. The broader matrix formulation of Harris, Iliopoulos, and Kolmogorov [HIK25] already permits selectors that remember past action labels. We apply this method to powered action probabilities and give a direct induction on execution prefixes. Corollary 3.5 identifies the information that ordinary matrices omit: $K_{f,1}$ does not determine $K_{f,s}$. Theorem 2.1 goes further: even every prefix power sum $Z_T(s)$ can miss an exponential improvement available by changing the repair order.

Achlioptas, Iliopoulos, and Vlassis [AIV17] use cuts at different depths of trajectory trees in their entropy-compression analysis of noisy local search. Our exact worst-case source interpretation follows from classical tree max-flow/min-cut and its use in Frostman’s lemma [BP17, Section 3.1]. The capped recursion also has the form of multiplicative optimal stopping: normalizing each positive row of R_s leaves its row sum as a multiplicative cost; see Jelito, Pitera, and Stettner [JPS21]. The scheduling separation and the explicit critical and composition formulas give quantitative consequences of this interpretation.

Our investigation of hierarchical covers was motivated by Li and Xu’s construction of local subspace priors combined with a prior over the Grassmannian [LX26, Section 3.3]. Xu’s Bellman-sufficient formulation studies which information must be retained to close a controlled recursion [Xu26]. In our setting, the controlled recursion optimizes prefix-cover capacity under a global constraint on tape probabilities.

The cube-root profile has a classical probabilistic analogue: critical branching Brownian motion with absorption has survival probability $\exp[-\Theta(t^{1/3})]$ for fixed initial position [BBS14]. Here the quantity is a maximum over globally constrained tape laws. Its proof uses the exact recursion $v \mapsto \min\{1, Rv\}$ and finite-dimensional spectral bounds. The local-lemma certificates and the disagreement examples avoid enumerating all assignments; the general controlled recursion need not admit such a reduction.

Harris and Srinivasan [HS17] bound Rényi entropy of the terminal Moser–Tardos distribution for enumerative applications. Our powered probabilities belong to surviving input-tape cylinders and describe continued execution. For extremal partial rejection, confluence and the probability-weighted trace series are established [CF69, GJL19, Jer24]; Abbes and Mairesse [AM15] characterize Bernoulli measures on infinite trace boundaries. The finite-state links between weighted growth, Hausdorff dimension, and Kolmogorov complexity are also classical [Sta93, Par64, Bow75, MW88, Sim15].

Reverse reconstruction is central to entropy compression [Mos09, MT10]. Messner and Thierauf [MT12] reconstruct bits used by satisfiability search and apply Kolmogorov complexity. Fialho, de Lima, and Procacci [FdLP22] show that the resampling and entropy-compression backtracking analyses they study yield identical bounds through the subset-gas cluster expansion. Appendix F gives a nonuniform backward likelihood identity, an arbitrary comparison law, and an exact relative-entropy formula for the slack. Our weak-source bounds analyze the specified search procedure directly; general simulation from weak randomness addresses a broader question [CG88, DMOZ23].

6 Further questions

The controlled recursion optimizes the repair order, but its state space can be exponential. Which local structures admit compact, exact or multiplicatively accurate representations of prefix-cover capacity? Can harmonic barriers extend beyond disagreement to overlapping repair systems? The component theorem also suggests seeking composition rules that retain a valid boundary potential and respect the global prefix constraint.

A Prefix covers and worst-case sources

Throughout this appendix, $P = p^{\mathbb{N}}$ is the full-support finite source from Section 1.2. The source class $\mathcal{Q}_{s,D}$ imposes its bound on every prefix of the full tape. Symbols after termination may be continued with the reference law.

Proof of Theorem 1.1. Every prefix cover gives $\nu(E) \leq D \sum_{w \in \mathcal{B}} P[w]^s$, which proves the upper bound. Suppose first that E is determined by T symbols. In the full source tree through depth T , give the root capacity $c(\emptyset) = 1$ and every other node w capacity $c(w) = DP[w]^s$. Route flow from the root only into the depth- T words whose cylinders lie in E . The tree max-flow/min-cut theorem gives a largest admissible flow f of value $\min\{1, DC_s(E)\}$. Indeed, cuts are exactly prefix covers, with root cost 1; a cover of a whole depth- T cylinder by longer words cannot lower its cost, since for nonnegative x_i and $s \leq 1$, $\sum_i x_i^s \geq (\sum_i x_i)^s$.

It remains to extend f to a probability law on all depth- T words. Extend f by zero on branches outside its support. For every internal node, including the root,

$$\sum_a c(wa) \geq c(w).$$

For nonroot nodes this follows from $\sum_a p(a)^s \geq 1$; at the root it also uses $D \geq 1$. Since f conserves flow, the residual capacities $r(w) = c(w) - f(w)$ satisfy

$$\sum_a r(wa) \geq r(w).$$

Start an additional flow g of value $1 - f(\emptyset)$ at the root. Whenever its incoming mass at w is at most $r(w)$, distribute that mass among its children without exceeding their residual capacities. The displayed inequality makes this possible at every step. Then $f + g$ has total mass one and respects all node capacities. It cannot give E more than the maximal value of f , because its restriction to successful leaves would itself be a larger admissible flow. Thus this probability law attains the upper bound. Continue it beyond depth T using the reference conditional law. For any suffix v ,

$$\nu[wv] = \nu[w]P[v] \leq DP[w]^s P[v]^s = DP[wv]^s,$$

so it belongs to $\mathcal{Q}_{s,D}$ on the infinite tape space.

For a closed E , let E_n be the union of the length- n cylinders that intersect E . These are decreasing closed-and-open sets with intersection E . Their contents decrease to $\mathcal{C}_s(E)$: monotonicity gives one inequality, while any finite prefix cover of E covers all sufficiently large E_n , by compactness. The class $\mathcal{Q}_{s,D}$ is nonempty and weakly compact. Indeed, $P \in \mathcal{Q}_{s,D}$, the tape space is compact, and each cylinder constraint is closed under weak convergence. Choose maximizers ν_n for E_n and a weakly convergent subsequence with limit $\nu_* \in \mathcal{Q}_{s,D}$. For fixed m and $n \geq m$,

$$\nu_n(E_m) \geq \nu_n(E_n) = \min\{1, DC_s(E_n)\}.$$

Pass to the subsequential limit, using that E_m is closed and open, and then let $m \rightarrow \infty$. Continuity from above yields $\nu_*(E) \geq \min\{1, DC_s(E)\}$, proving attainment and (5). For any decreasing sequence of finite-symbol events E_T with intersection $\mathcal{N}$, the same compactness argument shows $\mathcal{C}_s(E_T) \downarrow \mathcal{C}_s(\mathcal{N})$ and proves (6). $\square$

Corollary A.1 (Dimension as a source threshold). *Let $\mathcal{N}$ be closed and put $\delta = \dim_H^P \mathcal{N}$. For $0 < s < \delta$, some finite D and some $\nu \in \mathcal{Q}_{s,D}$ satisfy $\nu(\mathcal{N}) = 1$. For $\delta < s \leq 1$, every such law gives $\nu(\mathcal{N}) = 0$. For every $0 < s \leq 1$, the least domination constant permitting a law supported on $\mathcal{N}$ is*

$$\inf\{D \geq 1 : \exists \nu \in \mathcal{Q}_{s,D}, \nu(\mathcal{N}) = 1\} = \frac{1}{\mathcal{C}_s(\mathcal{N})}, \quad (42)$$

where $1/0 = \infty$. If $P(\mathcal{N}) = 0$, then

$$\delta = \inf\{s \in (0, 1] : \nu(\mathcal{N}) = 0 \text{ for every finite } D \geq 1 \text{ and every } \nu \in \mathcal{Q}_{s,D}\}. \quad (43)$$

At $s = \delta > 0$, there exists a law with finite domination constant supported on $\mathcal{N}$ exactly when $\mathcal{C}_\delta(\mathcal{N}) > 0$; dimension alone does not settle this endpoint.

Proof of Corollary A.1. In the metric $d_P(\omega, \eta) = P[\omega \wedge \eta]$, each cylinder $[w]$ has diameter $P[w]$. Any nonsingleton set is contained in a cylinder with the same diameter; singleton sets can be covered by cylinders of arbitrarily small diameter. Thus the infimum over countable cylinder covers equals the Hausdorff content. For compact $\mathcal{N}$, every such cover has a finite subcover, so this infimum is $\mathcal{C}_s(\mathcal{N})$. Its zero threshold is therefore $\dim_H^P \mathcal{N}$: the content is positive for $s < \delta$ and zero for $s > \delta$. Theorem 1.1 proves (42) and the strict-threshold claims. If $P(\mathcal{N}) = 0$, then $\mathcal{C}_1(\mathcal{N}) = P(\mathcal{N}) = 0$, so the set in (43) is nonempty and has infimum δ . The same exact identity gives the stated critical criterion. $\square$

B Proofs for scheduling and disagreement

The proofs follow the order of Sections 2 and 3. The finite-state dimension formula used below is proved separately in Appendix E.

B.1 Repair order and the prefix-cover recursion

Proof of Theorem 2.1. Put

$$u = \sqrt{\pi(0)} = \frac{\sqrt{15}}{4}, \quad v = \sqrt{\pi(1)} = \frac{1}{4}.$$

We first construct ν_n . Through time $8n$, generate actions with probabilities $3/4$ for 0 and $1/4$ for 1 until the n th zero appears; after that zero, output only zeros through time $8n$. After time $8n$, generate independent actions according to P . This specifies one law on tapes, without reference to the selector.

A prefix of length at most $8n$ preceding the n th zero, with k zeros, satisfies

$$\frac{\nu_n[w]}{P[w]^{1/2}} = \left(\frac{3}{\sqrt{15}}\right)^k \leq 1.$$

Suppose the n th zero appears at time $L \leq 8n$. For a positive-mass prefix of length $t \in [L, 8n]$, the corresponding ratio is

$$\left(\frac{3}{\sqrt{15}}\right)^n u^{-(t-L)} \leq \left(\frac{3/4}{u^8}\right)^n < 1,$$

since $L \geq n$ and $u^8 = (15/16)^4 > 3/4$. Zero-mass prefixes satisfy the required bound automatically. After time $8n$, each new reference action multiplies the ratio by $\pi(a)^{1/2} \leq 1$. Thus every finite

prefix satisfies the defining inequality for $\mathcal{Q}$. Under Λ_A , either fewer than n zeros have appeared by time $8n$, so an A task remains unfinished, or the B task is attempted only after the n th zero and subsequently sees zeros alone. In both cases $\tau > 8n$, proving (7).

For (8), consider an arbitrary $\nu \in \mathcal{Q}$ and use Λ_B . If no 1 appears during the first $4n$ actions, the tape begins with 0^{4n} , an event of ν -probability at most

$$u^{4n} = \left(\frac{225}{256}\right)^n.$$

Otherwise let $j \leq 4n$ be the first position containing 1. This completes B , and survival after $8n$ repairs requires fewer than n zeros among the remaining $\ell = 8n - j \geq 4n$ actions. Set $r = v/(3u) = 1/(3\sqrt{15})$. Their total power weight is at most

$$\begin{aligned} \sum_{k=0}^{n-1} \binom{\ell}{k} u^k v^{\ell-k} &\leq r^{-(n-1)} (v + ur)^\ell \\ &\leq r^{-(n-1)} (1/3)^{4n} = r \left(\frac{\sqrt{15}}{27}\right)^n. \end{aligned}$$

The first-1 prefix $0^{j-1}1$ has power weight $u^{j-1}v$. Applying the cylinder bound and summing over $j \leq 4n$ gives the claimed second term because

$$\sum_{j=1}^{4n} u^{j-1}v \leq \frac{v}{1-u} = 4 + \sqrt{15}.$$

This argument holds for every source in $\mathcal{Q}$, including the source ν_n used above. To obtain the simpler constant in (8), put $\alpha = \sqrt{15}/27$, $\beta = 225/256$, and $c = (4 + \sqrt{15})/(3\sqrt{15})$. The bound is $\beta^n + c\alpha^n$. Since $\alpha < \beta$ and $c\alpha/\beta = 256(4 + \sqrt{15})/18225 < 1/8$, it is at most $(9/8)\beta^n < 1$ for every $n \geq 1$.

For the lower bound in (8), take an i.i.d. source with probabilities u and $1 - u$. Both are at most the corresponding square-root reference probabilities, since $1 - u \leq v$. It belongs to $\mathcal{Q}$, and its prefix 0^{8n} leaves B unfinished under every selector. Its probability $u^{8n} = (15/16)^{4n}$ is therefore a lower bound for every selector, including the minimax optimum.

Finally, fix $s > 0$ and put

$$u_s = (15/16)^s, \quad v_s = (1/16)^s, \quad W_s = u_s + v_s.$$

Let P_s be the independent binary source with probabilities u_s/W_s and v_s/W_s . For every selector,

$$\tilde{Z}_T^\Lambda(s) = W_s^T \Pr_{P_s}^\Lambda\{\tau > T\}.$$

Under P_s , preassign an independent stream of actions to each task, and reveal its next action whenever that task is selected. Deferred revelation gives the same law as a fresh independent action at every repair, even when selection depends on all previously revealed labels. Each A task requires an independent geometric number of attempts with success probability u_s/W_s , and B requires an independent geometric number with success probability v_s/W_s . Every permitted selector makes exactly the sum of these attempts before all tasks complete. Consequently the stopping-time law under P_s , and hence $\tilde{Z}_T^\Lambda(s)$, is independent of Λ .

For clarity about the indexing convention, the completed- T definition includes a repair that terminates the computation at time T . Thus $Z_0^\Lambda(s) = 1$ and, for $T \geq 1$,

$$Z_T^\Lambda(s) = W_s^T \Pr_{P_s}^\Lambda\{\tau \geq T\} = W_s \tilde{Z}_{T-1}^\Lambda(s).$$

Its selector invariance follows as well. Taking $s = 1$ proves the stated ordinary stopping-time law. Under any deterministic selector, an infinite run has a last task completion, since there are only $n + 1$ tasks. After that finite history each chosen task must receive its unique failure symbol. The selector therefore determines the rest of the tape from that history. There are only countably many finite histories, so the nonterminating set is countable and has Hausdorff dimension zero. The all-prefix bound makes every $\nu \in \mathcal{Q}$ atomless, hence every such selector terminates almost surely. $\square$

Proof of Proposition 2.2. Fix a finite horizon. At a live root, a prefix cover either cuts the root at cost one or covers the children after the chosen repair. For a specified first repair f , the selector can choose its continuation independently after each observed symbol a . The optimal continuation cost is $V_t(\Phi_f(x, a))$, multiplied by the edge cost $\pi(a)^s$. Minimizing over f and comparing with the root cut gives (9) by induction. At the horizon, the surviving event costs one at a live state and zero at a terminated state, exactly the stated boundary values. Each minimum is attained, and an attaining choice can depend only on current state and remaining time. These choices depend on s but not on D , and the same state-based rule is available from every initial state. Theorem 1.1 now turns the optimal cover cost into (10), since $c \mapsto \min\{1, Dc\}$ is nondecreasing.

This proof optimizes one globally constrained source. It does not reset the domination constant after a prefix or give the source independent conditional choices at each step. The selector has no additional private randomness; allowing trusted private coins would define a different optimization problem. $\square$

B.2 Stopping times and dimension for disagreement

Proof of Theorem 3.1. Assign an increment $d(a) \in \{-1, 0, 1\}$ to each action class and put $Y_t = k_0 + \sum_{i=1}^t d(\omega_i)$. For every tape and selector,

$$\tau(\omega) = \inf\{t \geq 0 : Y_t \in \{0, N\}\}, \quad \inf \emptyset = \infty. \quad (44)$$

Until termination the selected edge contains one zero and one one, so the actual Hamming weight equals Y_t . Connectivity says that exactly weights zero and N are terminal. Hence the entire stopping time as a function of the tape, and its nonterminating set, depend only on N , k_0 , and the action partition. The graph and selector do not enter. The three increments have total powered weights D_s, N_s, U_s , giving $e_{k_0}^\top R_s^j \mathbf{1}$ for remaining live after j repairs. Completing $T \geq 1$ repairs permits any final action after $T - 1$ live transitions, proving (14).

The matrix R_s is diagonally similar to the symmetric tridiagonal matrix with diagonal N_s and off-diagonal $(D_s U_s)^{1/2}$. Its eigenvalues are

$$N_s + 2(D_s U_s)^{1/2} \cos \frac{j\pi}{N}, \quad j = 1, \dots, N - 1.$$

This proves (15). At $s = 1$,

$$\varrho_1 = N_1 + 2\sqrt{D_1 U_1} c_N < N_1 + D_1 + U_1 = 1.$$

As $s \downarrow 0$, $D_s \rightarrow |\mathcal{A}_-|$, $N_s \rightarrow |\mathcal{A}_p| + |\mathcal{A}_r|$, and $U_s \rightarrow |\mathcal{A}_+|$. Hence $\varrho_s > 1$ near zero. Every source atom lies strictly between zero and one, so ϱ_s is continuous and strictly decreasing. The process therefore terminates exponentially fast and the critical power Δ is unique.

For the lower bound matching the dimension upper estimate, put

$$h_s(k) = \left(\frac{D_s}{U_s}\right)^{k/2} \sin \frac{k\pi}{N}, \quad 1 \leq k < N,$$

and set $h_s(0) = h_s(N) = 0$. Then $R_s h_s = \varrho_s h_s$. At $s = \Delta$, if action a has reference probability $\pi(a)$ and changes the Hamming weight by $d(a) \in \{-1, 0, 1\}$, define

$$\nu_\Delta(a \mid k, z) = \pi(a)^\Delta \frac{h_\Delta(k + d(a))}{h_\Delta(k)}. \quad (45)$$

The eigenvector equation says that these probabilities sum to one. Boundary moves receive probability zero, so ν_Δ is supported on $\mathcal{N}$. For every prefix w in its support, ending at weight k_T ,

$$\nu_\Delta[w] = P[w]^\Delta \frac{h_\Delta(k_T)}{h_\Delta(k_0)}. \quad (46)$$

The ratio is bounded above and below on $\{1, \dots, N-1\}$. The source-relative mass-distribution argument gives the lower Hausdorff bound; (14) at every $s > \Delta$ gives the upper bound.

If π is computable, the strictly decreasing critical equation makes Δ computable, and the displayed Perron law is computable on the finite Hamming-weight automaton. A tape random for this law lies in $\mathcal{N}$ and has effective strong dimension at least Δ by Levin–Schnorr and (46). Conversely, the live-prefix family is computable from the increments, independently of the selector. The critical powered sums are uniformly bounded because R_Δ is similar to a symmetric matrix of spectral radius one. Coding the length- T live prefixes with weights proportional to $P[w]^\Delta$, and encoding T , gives

$$K(w) \leq \Delta(-\log_2 P[w]) + O(\log(T+1)).$$

Since $-\log_2 P[w] = \Theta(T)$, this proves the reverse bound and attainment without any computability assumption on the selector.

Finally, if (18) holds, disjointness of the live cylinders and (14) give (19). Equation (46) gives the required critical law and domination constant. Thus the same Δ is the exact global prefix-domination threshold and proves all remaining claims. $\square$

B.3 The exact robustness recursion

Proof of Theorem 3.2. Fix a horizon T and a consumed prefix w ending at a live assignment. A prefix-free cover of its surviving continuations either contains w itself or covers the relevant descendants separately. Divide its cost by $P[w]^s$. The first option costs one; the second has cost equal to the sum of the normalized child costs, weighted by $\pi(a)^s$. At horizon zero, the cost is one at a live state and zero at a terminal state. Backward induction therefore gives (20): the total powered masses of actions decreasing, preserving, and increasing Hamming weight are D_s, N_s, U_s , regardless of the selected flawed edge or the earlier history. Theorem 1.1 converts this minimum cover cost into (21), including attainment. Tridiagonality gives the arithmetic and storage bounds.

The map $x \mapsto \min\{1, R_s x\}$ is monotone and maps $[0, 1]^{N-1}$ into itself. Its iterates from $\mathbf{1}$ decrease to the greatest fixed point x_s in this cube. The decreasing-event conclusion of Theorem 1.1 gives

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu(\mathcal{N}) = \min\{1, Dx_s(k_0)\}.$$

For completeness, $x_s = 0$ when $\rho(R_s) < 1$, since $C_{T,s} \leq R_s^T \mathbf{1}$. If $\rho(R_s) \geq 1$, a positive Perron vector normalized to have maximum one is a subsolution; induction shows that it lies below every $C_{T,s}$. Hence x_s is positive.

At $s = \Delta$, let $\ell > 0$ be a left Perron vector of R_Δ . The fixed-point equation implies $x_\Delta \leq R_\Delta x_\Delta$, but

$$\ell^\top (R_\Delta x_\Delta - x_\Delta) = 0.$$

Every coordinate of the difference is nonnegative, so positivity of ℓ forces $x_\Delta = R_\Delta x_\Delta$. Irreducibility makes this vector a multiple of the positive right Perron vector $h(k) = (D_\Delta/U_\Delta)^{k/2} \sin(k\pi/N)$. The greatest such vector bounded by one is $h/h_{\max}$. This proves (22) and the optimal domination constant. The critical Perron law constructed in Appendix B.2 attains that constant. $\square$

B.4 Matching finite-time bounds and the gap from single-depth covers

For the proof, fix $N \geq 3$ and put

$$\theta = \frac{\pi}{N}, \quad r = 7^{\Delta_N}, \quad a = (7/100)^{\Delta_N}, \quad b = 2(1/4)^{\Delta_N}.$$

Then $b + 2a \cos \theta = 1$. For $2 \leq m \leq N$, let $R^{(m)}$ be the tridiagonal matrix on $1, \dots, m-1$ with lower diagonal ar , diagonal b , and upper diagonal a/r , and set

$$L_m = \text{diag}(r, \dots, r^{m-1}), \quad \lambda_m = b + 2a \cos(\pi/m).$$

Thus $R^{(m)} = L_m(bI + aA_{m-1})L_m^{-1}$, where A_{m-1} is path adjacency. We write $R = R^{(N)}$.

Proof of the finite-time bounds (24). The critical equation gives $1/2 < \Delta_N < 1$. Consequently

$$\sqrt{7} < r < 7, \quad 7/100 < a < \sqrt{7}/10, \quad b > 1/2, \quad b > 2a.$$

In particular, the symmetric matrix $bI + aA_{m-1}$ is positive definite with operator norm $\lambda_m \in (1/2, 1]$, and $\lambda_N = 1$. For any fixed initial assignment with k ones, the exact prefix-cover formula identifies $\max_{\nu \in \mathcal{Q}_{\Delta_N, 1}} \mathbb{P}_\nu\{\tau > t\}$ with $v_t(k) := C_{t, \Delta_N}(k)$. These values obey

$$v_0 = \mathbf{1}, \quad v_{t+1} = \min\{\mathbf{1}, Rv_t\}.$$

They are independent of the assignment and selector. Put $q = \min\{T^{1/3}, N\}$. All constants below are absolute.

For the lower bound, normalize the Perron vector on $1, \dots, m-1$:

$$g_m(k) = \frac{r^k \sin(k\pi/m)}{r^{m-1} \sin(\pi/m)} \quad (k < m), \quad g_m(k) = 0 \quad (k \geq m).$$

This vector lies in $[0, 1]^{N-1}$. Indeed, the sine recurrence gives $\sin(k\pi/m) \leq 2 \cos(\pi/m) \sin((k+1)\pi/m)$ for $1 \leq k \leq m-2$, and $r > 2$ makes the numerator increasing. Since $Rg_m \geq \lambda_m g_m$, induction in the truncated recursion gives the first barrier bound:

$$v_T(1) \geq r^{-(m-2)} \lambda_m^T. \tag{47}$$

Moreover,

$$-\log \lambda_m \leq 2(1 - \lambda_m) = 4a(\cos \theta - \cos(\pi/m)) = O(m^{-2}).$$

Choose $m = \min\{N, \max\{2, \lceil T^{1/3} \rceil\}\}$. If $m = N$, then $\lambda_m = 1$ and $m - 2 \leq q$. Otherwise $m - 2 \leq q$ and $T/m^2 \leq q$. Thus (47) gives $v_T(1) \geq e^{-Cq}$.

For the upper bound, take $2 \leq m \leq N/2$ and write $h(k) = r^k \sin(k\theta)$ and $z(k) = h(k)/h(m)$ for $0 \leq k \leq m$. The same sine recurrence shows that h is increasing. Also $Rh = h$, so z is harmonic on $1, \dots, m-1$, with boundary values zero and one. Run the truncated recursion from one on those interior weights, keeping the two boundary values fixed, and denote its values by w_t . Monotonicity gives $v_t \leq w_t$ there because $v_t(m) \leq 1$. Since z is a fixed point and $w_0 \geq z$, the nonnegative difference satisfies $w_{t+1} - z \leq R^{(m)}(w_t - z)$ on the interior. The diagonal similarity and the Euclidean norm therefore give

$$w_T(1) - z(1) \leq r \lambda_m^T \|L_m^{-1}(w_0 - z)\|_2 \leq \frac{r}{\sqrt{r^2 - 1}} \lambda_m^T.$$

This is the second barrier bound:

$$v_T(1) \leq r^{1-m} \frac{\sin \theta}{\sin(m\theta)} + \frac{r}{\sqrt{r^2 - 1}} \lambda_m^T. \quad (48)$$

For $m \leq N/2$, elementary sine bounds give, uniformly,

$$\frac{\sin \theta}{\sin(m\theta)} \leq \frac{\pi}{2m}, \quad 1 - \lambda_m = 2a \int_{\pi/N}^{\pi/m} \sin x \, dx = \Omega(m^{-2}).$$

Hence the right side of (48) is at most $C_1(e^{-c_1 m} + e^{-c_1 T/m^2})$. For $N \geq 4$ and $T \geq 8$, choose $m = \lfloor \min\{T^{1/3}, N/2\} \rfloor$. Then $m \geq q/4$ and $T/m^2 \geq q$, giving $v_T(1) \leq C_2 e^{-c_2 q}$. If $N = 3$ or $T < 8$, then $q \leq 3$; increasing C_2 covers these cases.

Finally, v_t decreases with t , and

$$v_1(1) = b + a/r = 1 - a(2 \cos \theta - 1/r) \leq 1 - \frac{7}{100}(1 - 1/\sqrt{7}) < 1.$$

This uniform strict bound absorbs the prefactor C_2 when q is bounded; when q is sufficiently large it is absorbed by decreasing the exponential rate. Thus $v_T(1) \leq e^{-cq}$ with an absolute $c > 0$. Taking $T = N^3$ proves sufficiency of the cubic horizon. Conversely, $v_T(1) \leq e^{-\beta N}$ for any fixed $\beta > 0$ forces $T \geq (\beta/C)^3 N^3$ by the lower bound. This proves both claims. $\square$

The limit and the single-depth gap. We now prove (25) and (26), together with the following relative-accuracy refinement: at $T_N = 3N^4$,

$$C_{T_N, \Delta_N}(1) \leq 2 \cdot 7^{-\Delta_N(N-2)}. \quad (49)$$

This is stronger than merely reaching $\exp[-\Theta(N)]$, whose sharp time scale is N^3 .

Proof. Put $s = \Delta_N$ and $S = bI + aA_{N-1}$. The symmetric matrix S has eigenvalues $\mu_j = b + 2a \cos(j\theta)$ and orthonormal eigenvectors $u_j(k) = \sqrt{2/N} \sin(jk\theta)$; the similarity is $R = L_N S L_N^{-1}$. Here $\mu_1 = 1$, and all eigenvalues are positive because $b > 2a$.

The right Perron vector $h(k) = r^k \sin(k\theta)$ is strictly increasing, as shown in the finite-time proof. Theorem 3.2 now yields

$$C_{\infty, s}(1) = \frac{h(1)}{h(N-1)} = r^{-(N-2)},$$

which proves (25).

For the finite horizon, write $v_t = C_{t,s}$ and $v_* = h/h(N-1)$. Since $v_t \downarrow v_*$ and $v_*(N-1) = 1$, the last coordinate of v_t equals one at every time. Thus the nonnegative error $e_t = v_t - v_*$ has last coordinate zero. On the remaining coordinates,

$$e_{t+1} \leq R^- e_t, \quad 0 \leq e_0 \leq \mathbf{1},$$

where R^- is the principal submatrix on $1, \dots, N-2$. Indeed, $Rv_* = v_*$ and truncating Rv_t at one can only decrease the error. With $L^- = \text{diag}(r, \dots, r^{N-2})$,

$$R^- = L^- S^- (L^-)^{-1}, \quad S^- = bI + aA_{N-2}.$$

The symmetric matrix S^- is positive definite, with largest eigenvalue

$$\lambda_* = 1 - 2a \left(\cos \frac{\pi}{N} - \cos \frac{\pi}{N-1} \right) \in (0, 1). \quad (50)$$

Its Euclidean operator norm gives

$$\begin{aligned} e_t(1) &\leq r \lambda_*^t \|(L^-)^{-1} e_0\|_2 \leq r \lambda_*^t \left(\sum_{k=1}^{N-2} r^{-2k} \right)^{1/2} \\ &\leq \sqrt{\frac{7}{6}} \lambda_*^t. \end{aligned}$$

The bound $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ implies

$$\cos \frac{\pi}{N} - \cos \frac{\pi}{N-1} \geq \frac{\pi(2N-1)}{N^2(N-1)^2} \geq \frac{2\pi}{N^3}, \quad 1 - \lambda_* \geq \frac{7\pi}{25N^3}.$$

At $T_N = 3N^4$, therefore,

$$e_{T_N}(1) \leq \sqrt{\frac{7}{6}} \exp \left(-\frac{21\pi}{25} N \right) \leq 7^{-(N-2)} \leq r^{-(N-2)}.$$

This proves (49).

Finally, nonnegativity and the positive spectrum of S give, for every $t \geq 0$,

$$\tilde{Z}_t(s) \geq (R^t)_{11} = (S^t)_{11} = \sum_{j=1}^{N-1} \mu_j^t u_j(1)^2 \geq \frac{2}{N} \sin^2 \frac{\pi}{N} \geq \frac{8}{N^3}.$$

For the matching order, the spectral expansion also gives

$$\lim_{t \rightarrow \infty} \tilde{Z}_t(s) = \frac{2r}{N} \sin \theta \sum_{k=1}^{N-1} r^{-k} \sin(k\theta) \leq \frac{2\pi^2}{N^3} \left(\frac{r}{r-1} \right)^2 = O(N^{-3}),$$

using $\sin(k\theta) \leq k\theta$ and $r > \sqrt{7}$. Thus $\inf_t \tilde{Z}_t(s) = \Theta(N^{-3})$ uniformly in N . At $T_N = 3N^4$, the finite-horizon optimum lies between $r^{-(N-2)}$ and twice this quantity, and so is exponentially small. All reductions used only the Hamming weight, making the conclusions uniform over graphs and selectors. $\square$

B.5 Exact composition of disagreement components

Proof of Theorem 3.4. Write $s = \Delta_N$, $a = (7/100)^s$, $b = 2(1/4)^s$, $d = (49/100)^s = ar$, and $u = (1/100)^s = a/r$. Thus

$$b + 2a \cos \theta = 1, \quad r > \sqrt{7} > 2 \cos \theta, \quad b + d > 1.$$

The function $h(k) = r^k \sin(k\theta)$ is strictly increasing for $1 \leq k \leq N-1$ and obeys $dh(k-1) + bh(k) + uh(k+1) = h(k)$ for $1 \leq k < N$, where $h(0) = h(N) = 0$. Indeed, $\sin(k\theta) \leq 2 \cos \theta \sin((k+1)\theta)$ for $1 \leq k \leq N-2$, so $h(k+1) > h(k)$.

At any history let K count ones in the components that are still nonconstant. A downward action sends K to $K-1$, and either neutral action preserves K . An upward action normally sends K to $K+1$. If it completes an L -vertex component, however, it sends K to $K+1-L$. Such a completion is possible only when $K \geq L-1 \geq N-1$. This dependence on the selected component prevents a birth-death reduction of the entire transition law.

Put $V = V_N$. We first prove the upper bound. Only $1 \leq K_0 < N-1$ needs consideration, since $V(K_0) = 1$ above that range. Cover a nonterminating tape at its first visit to $K = N-1$, or at depth T if no such visit has occurred. Before this visit no component can complete upward, so the powered count process is the tridiagonal matrix A on $1, \dots, N-2$, with lower diagonal d , diagonal b , and upper diagonal u . Its spectral radius is

$$\rho(A) = b + 2a \cos \frac{\pi}{N-1} < 1.$$

The harmonic identity, with boundary values zero at 0 and one at $N-1$, says that the total powered mass of first visits to $N-1$ is $V(K_0)$. Survivors below that level at depth T have powered mass $e_{K_0}^T A^T \mathbf{1} \rightarrow 0$. These finite covers therefore have costs at most $V(K_0) + e_{K_0}^T A^T \mathbf{1}$. Theorem 1.1 gives the desired upper bound $\min\{1, DV(K_0)\}$.

We construct one source attaining this bound for all allowed systems. Suppose $K_0 > 0$, and let an auxiliary law q maintain a counter $Y_0 = K_0$. For $1 \leq Y_t < N-1$, set

$$q(a \mid Y_t) = \pi(a)^s \frac{V(Y_t + d(a))}{V(Y_t)},$$

where $d(a) \in \{-1, 0, 1\}$ is the action increment. For $Y_t \geq N-1$, give each neutral action probability $(1/4)^s$, the downward action probability $1-b$, and the upward action probability zero. In both cases update $Y_{t+1} = Y_t + d(a)$. The harmonic identity normalizes the probabilities below $N-1$; at weight one the downward probability is zero. Above that range normalization is immediate. Hence Y_t never reaches zero.

Every upward move starts below $N-1$, so it cannot complete a component of size at least N . Downward completion removes a zero component and has no additional effect on the count. Thus the actual unresolved-one count equals Y_t on every tape generated by q , for every graph, initial assignment, and selector under consideration. All these tapes are nonterminating. Moreover,

$$dV(N-2) + b = 1, \quad d + b > 1$$

give, in the upper range as well as the lower range,

$$q(a \mid Y_t) \leq \pi(a)^s \frac{V(Y_{t+1})}{V(Y_t)}.$$

Multiplying along a prefix yields

$$q[w] \leq P[w]^s \frac{V(Y_{|w|})}{V(K_0)}. \quad (51)$$

Put $p = \min\{1, DV(K_0)\}$ and $\eta = pq$. This subprobability law has root mass $p \leq 1$ and satisfies $\eta[w] \leq DP[w]^s$. Apply the residual-capacity construction of Appendix A recursively at every node of the infinite source tree, in a fixed action order. It extends η to a probability law $\nu \in \mathcal{Q}_{s,D}$. This construction uses only N, K_0, D , not the graph or selector. Every allowed process has nontermination probability at least p under ν , and the upper bound gives equality. If $K_0 = 0$, all components are already constant and P itself is a common maximizer. In particular, when $K_0 \geq N - 1$ and $D = 1$, the law q needs no completion and forces nontermination under every interleaving.

Finally, under P the usual downward drift is $49/100 - 1/100 = 12/25$. Completing an all-one component only decreases K further, so at every live history $\mathbb{E}[K_{t+1} - K_t \mid \text{history}] \leq -12/25$. Summing this inequality up to $\tau \wedge t$, using $K \geq 0$, and then letting $t \rightarrow \infty$ proves (28). $\square$

Corollary B.1 (A sharp failure of robustness under composition). *Take M connected components, each of size at least N , each initially with one vertex equal to one. For every selector,*

$$\max_{\nu \in \mathcal{Q}_{\Delta_N, 1}} \mathbb{P}_\nu\{\tau = \infty\} = \begin{cases} r^{M-N+1} \frac{\sin(M\pi/N)}{\sin(\pi/N)}, & 1 \leq M < N - 1, \\ 1, & M \geq N - 1. \end{cases}$$

Each component in isolation has capacity $r^{-(N-2)}$ when its size is N (and the same capacity for every larger size). Yet $N - 1$ such components permit certain nontermination under the same global prefix constraint. For all $1 \leq M \leq \lfloor N/2 \rfloor$, the joint capacity is $\exp[-\Theta(N)]$, with absolute constants, whereas $\mathbb{E}_P \tau \leq 25M/12$.

If all components have size N , every single-depth surviving-prefix sum at exponent Δ_N , for every selector, is at least $2N^{-1} \sin^2(\pi/N)$. Thus the exponential separation from every single-depth cover persists for $\Theta(N)$ modules on $\Theta(N^2)$ variables.

Proof. Substitute $K_0 = M$ in the theorem. Strict increase of h gives a value strictly below one whenever $M < N - 1$, so the threshold is exact. The exponential estimate follows from $\sqrt{7} < r < 7$ and

$$r^{-(N-2)} \leq V_N(M) \leq \frac{N}{2} r^{-(N/2-1)} \quad (1 \leq M \leq \lfloor N/2 \rfloor),$$

absorbing finitely many small N into the absolute constants. Let $W_s = \sum_a \pi_a^s$ and give action a tilted probability π_a^s / W_s . The full surviving-prefix sum is W_s^t times the probability of survival after t steps under this i.i.d. tilted source. Preassign an independent action stream to each component and reveal the next symbol of its stream whenever it is selected. For each stream, the local repair count is determined by its weight increments, regardless of the internal edge choices or any external history. Consequently the total repair count is the sum of independent local repair counts, and its distribution is independent of the interleaving selector. It is at least the first component's repair count. The corresponding powered sum is therefore at least $e_1^T R_s^t \mathbf{1}$, which is at least $2N^{-1} \sin^2(\pi/N)$ by (26). $\square$

B.6 The same transition matrices with different dimensions

Here is the construction. Choose an integer $m \geq 2$, a number $0 < t < 1/2$, and put $q = 1/2 - t$. The common alphabet is $\{x_1, \dots, x_m, y, u, v\}$, with

$$\pi(x_i) = q/m, \quad \pi(y) = q, \quad \pi(u) = \pi(v) = t. \quad (52)$$

For a selected edge reading z , use the two maps

$$\begin{array}{c|cccc} & x_1, \dots, x_m & y & u & v \\ \hline H & z & 00 & 11 & \bar{z} \\ L & 00 & z & 11 & \bar{z}. \end{array} \quad (53)$$

Both maps therefore assign ordinary masses (q, q, t, t) to $(00, z, 11, \bar{z})$, so their $K_{f,1}$ matrices agree. The labels $x_1, \dots, x_m$, however, collide at one transition. For $J \in \{H, L\}$, let $\varrho_J(s) = \lim_{T \rightarrow \infty} Z_T^J(s)^{1/T}$. The exact calculation below shows that dimension is the unique root $\varrho_J(s) = 1$. On P_4 the rates are

$$\varrho_H(s) = m^{1-s} q^s + t^s + \sqrt{2} (qt)^{s/2}, \quad (54)$$

$$\varrho_L(s) = q^s + t^s + \sqrt{2} m^{(1-s)/2} (qt)^{s/2}. \quad (55)$$

As $t \downarrow 0$, their roots converge, respectively, to

$$\frac{\log m}{\log(2m)} \quad \text{and} \quad 0. \quad (56)$$

Taking m large and then t small proves (29). The languages themselves differ: from Hamming weight one, x_1^∞ survives under H and stops under L , while y^∞ does the reverse. The complete proof follows below.

The common imperfect source can be chosen explicitly:

$$\nu_0 = \text{Unif}\{x_1, \dots, x_m\}^{\mathbb{N}}, \quad \beta = \frac{\log m}{\log(m/q)}. \quad (57)$$

For a word w of length T using only these labels,

$$\nu_0[w] = m^{-T} = P[w]^\beta;$$

for every other word, $\nu_0[w] = 0$. Under H these labels leave the state unchanged. Under L each label reduces Hamming weight by one, so the process stops after exactly k_0 repairs. Both assertions are independent of the selector. As $t \downarrow 0$, $\beta \rightarrow \log m / \log(2m)$, while $\Delta_L \rightarrow 0$. Also $\varrho_H(\beta) > 1$, so $\beta < \Delta_H$. Taking m large and then t small permits $s_0 = \beta > 1 - \varepsilon$ in the corollary.

Let $K_{f,s}(\sigma, \tau)$ sum the s th powers of action probabilities producing the transition $\sigma \rightarrow \tau$ when flaw f is repaired. The role of collisions then has a simple information-theoretic form. If $r = K_{f,1}(\sigma, \tau) > 0$ and χ is the conditional law of the action label among those producing $\sigma \rightarrow \tau$, then

$$K_{f,s}(\sigma, \tau) = r^s \sum_a \chi(a)^s = r^s 2^{(1-s)H_s(\chi)}. \quad (58)$$

Here H_s is base-two Rényi entropy, with its continuous interpretation at $s = 1$. Thus $K_{f,1}$ records the total transition mass, whereas the source powers also record the Rényi entropy hidden behind that transition. Under action recoverability every fiber is a singleton and this extra term vanishes. In (53), an m -way collision moves between the neutral and downward transitions while all power-one data remain fixed.

Action splitting can already change dimension in a one-state loop. The role of this example is to realize that loss of information with overlapping local repairs and a nontrivial commuting pair, uniformly over all selectors.

Adjacent edge flaws on P_4 overlap and can occur simultaneously, whereas the two end-edge repairs have disjoint scopes and commute at every source power. Thus commutation is nonvacuous,

but the example is genuinely nonextremal and overlapping. The example shows that even the entire ordinary repair kernel can lose essentially all information about weak-source robustness and nontermination dimension.

Proof of Corollary 3.5. For map H in (53),

$$D_s^H = q^s, \quad N_s^H = m^{1-s}q^s + t^s, \quad U_s^H = t^s.$$

For map L ,

$$D_s^L = m^{1-s}q^s, \quad N_s^L = q^s + t^s, \quad U_s^L = t^s.$$

Since $c_4 = 1/\sqrt{2}$, Theorem 3.1 gives (54) and (55). At $s = 1$ both rules assign masses (q, q, t, t) to $(00, z, 11, \bar{z})$ for every current oriented word z . Thus $K_{f,1}^H = K_{f,1}^L$ for every flaw and state. Their common matrix R_1 also makes the complete stopping-time law identical for every selector by (14); this remains true when the selector sees action labels, because the recurrence is independent of the selected edge and the earlier labels.

For every fixed $s > 0$, as $t \downarrow 0$ one has

$$\varrho_H(s) \longrightarrow m^{1-s}2^{-s}, \quad \varrho_L(s) \longrightarrow 2^{-s}.$$

The first limiting function has unique root $\log m / \log(2m)$. Strict monotonicity and continuity therefore give the first limit in (56). For the second, fix any $s_0 > 0$. Since $2^{-s_0} < 1$, one has $\varrho_L(s_0) < 1$ for all sufficiently small t , and hence $\Delta_L < s_0$. Thus $\Delta_L \rightarrow 0$. Choose m so that $\log m / \log(2m) > 1 - \varepsilon$, and then choose a positive rational t small enough to obtain (29) and $\beta > \max\{\Delta_L, 1 - \varepsilon\}$, where β is defined in (57). The source and both maps are then computable, so Theorem 3.1 also gives attainment of the maximal effective strong dimensions.

For the operational assertion, take ν_0 and β from (57). A word of length T consisting only of x_i symbols has probabilities m^{-T} under ν_0 and $(q/m)^T$ under P ; hence $\nu_0[w] = P[w]^\beta$. All other words have ν_0 -probability zero. The identity $m^{1-\beta}q^\beta = 1$ implies

$$\varrho_H(\beta) = 1 + t^\beta + \sqrt{2}(qt)^{\beta/2} > 1,$$

and therefore $\beta < \Delta_H$. The parameter choices give $\beta > \max\{\Delta_L, 1 - \varepsilon\}$, so take $s_0 = \beta$. Every x_i action is neutral under H and downward under L . Thus H never stops, and L stops after exactly k_0 repairs, for every initial live assignment and every pair of selectors. This proves the claimed separation under a common independent source. $\square$

B.7 Conditional action bounds in disagreement repair

Proposition B.2 (An exact one-step source condition). *Suppose $N \geq 3$ and $0 < s \leq 1$. At each positive-probability live history, allow any action distribution satisfying*

$$\mathbb{P}_\nu\{A_t = a \mid \mathcal{H}_{t-1}\} \leq \pi(a)^s \quad (a \in \mathcal{A}). \quad (59)$$

If $N_s + \min\{D_s, U_s\} < 1$, every such source has an exponential stopping tail, uniformly over all deterministic nonanticipating selectors. More precisely, with $\eta = 1 - N_s - \min\{D_s, U_s\} > 0$,

$$\mathbb{P}_\nu\{\tau > j(N-1)\} \leq (1 - \eta^{N-1})^j \quad (j \geq 0). \quad (60)$$

If $N_s + \min\{D_s, U_s\} \geq 1$, then, for each live initial Hamming weight, there is a source satisfying (59) on which the process never terminates, for every selector. One rule for choosing the next action, depending only on the current weight, applies to all initial assignments; the induced tape law may depend on the initial weight.

Proof. Suppose first that $U_s \leq D_s$ and $N_s + U_s < 1$. The total conditional probability of neutral or upward actions is at most $N_s + U_s$, so a downward action has conditional probability at least $\eta = 1 - N_s - U_s$. Starting from any live weight, $N - 1$ consecutive downward actions force termination. The conditional probability of termination within the next $N - 1$ repairs is therefore at least η^{N-1} . Applying this bound at successive blocks proves (60). If $D_s \leq U_s$, use upward actions instead.

For the converse, both $N_s + U_s$ and $N_s + D_s$ are at least one. At weight one, permit only neutral and upward actions; at weight $N - 1$, permit only neutral and downward actions; at interior weights permit all actions. In every case the sum of the bounds $\pi(a)^s$ over permitted actions is at least one. Choose probabilities summing to one within those bounds, for example by filling them successively in a fixed action order. The resulting law satisfies (59) and never leaves $\{1, \dots, N - 1\}$. Since weight changes depend only on the action class, the construction is independent of the selector. $\square$

The two boundaries differ in general. The dimension root satisfies

$$N_s + 2\sqrt{D_s U_s} \cos(\pi/N) = 1.$$

A conditional action bound implies global cylinder domination, but the latter permits the source to shift randomness between different times.

C A local-lemma bound for weighted prefixes

This section bounds the complete prefix power sum of an overlapping repair process without constructing its full assignment chain. The input is a family of powered transition matrices and local scalar bounds on their growth. The conclusion is uniform over the order in which present flaws are selected.

Let Ω be a finite state space and let $F_\sigma \subseteq F$ be the flaws present at σ . Addressing $f \in F_\sigma$ chooses an action $a \in \mathcal{A}_f(\sigma)$ with probability $\rho_f(a \mid \sigma) > 0$ and moves to $\Phi_f(\sigma, a)$. The action sets are finite and satisfy

$$\sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma) = 1.$$

Distinct actions may lead to the same terminal state. A deterministic nonanticipating selector may use the complete observed past, including past action labels, but not future randomness.

Let $\sim$ be a symmetric relation on distinct flaws and define the inclusive neighborhood

$$\Gamma(f) = \{f\} \cup \{g \in F : g \sim f\}.$$

In local-repair applications, $\sim$ is also chosen to satisfy the potential-causality condition

$$F_{\Phi_f(\sigma, a)} \subseteq (F_\sigma \setminus \{f\}) \cup \Gamma(f). \quad (61)$$

This is Kolmogorov's convention after adding a self-loop at every flaw. The matrix theorem itself uses only powered commutation; potential causality explains why D is a local dependency graph. Write $\text{Ind}(S)$ for the subsets of S whose distinct elements are pairwise nonadjacent.

For $s > 0$, define the powered action kernel

$$K_{f,s}(\sigma, \tau) = \mathbf{1}\{f \in F_\sigma\} \sum_{\substack{a \in \mathcal{A}_f(\sigma) \\ \Phi_f(\sigma, a) = \tau}} \rho_f(a \mid \sigma)^s. \quad (62)$$

The kernel aggregates actions with the same endpoints, but for $s \neq 1$ its entries retain the powered probabilities of the individual actions and need not be determined by the ordinary transition matrix $K_{f,1}$. The process is *s-commutative* if

$$K_{f,s}K_{g,s} = K_{g,s}K_{f,s} \quad (f \neq g, f \not\sim g). \quad (63)$$

For independent redraws on disjoint scopes, swapping the two actions preserves their endpoints and probability product, so commutation holds at every power. With exactly M_f equiprobable actions for flaw f , $K_{f,s} = M_f^{1-s}K_{f,1}$, and ordinary commutation is enough. Lemma C.4 gives the general sufficient condition. Ordinary commutation alone need not survive powering when action probabilities are nonuniform.

Fix an initial law u on Ω . The initial source symbol is taken to be the initial state, and its powered mass vector is

$$b_s(\sigma) = u(\sigma)^s.$$

If instead an initial symbol α of probability $q(\alpha)$ produces the state $\iota(\alpha)$, replace this by

$$b_s(\sigma) = \sum_{\alpha: \iota(\alpha)=\sigma} q(\alpha)^s.$$

One later source symbol is one complete oracle action. Let $\mathcal{L}_T^\Lambda$ be the prefix-free family of minimal tape prefixes that complete T repairs under selector Λ , and set

$$Z_T^\Lambda(s) = \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s, \quad (64)$$

where $P_{\text{tape}}[w]$ includes the initial source symbol. With this convention, $Z_T^\Lambda(1) = \mathbb{P}\{\tau \geq T\}$ when τ is the number of completed repairs.

Choose a strictly positive probability vector v_s on Ω and define

$$\lambda_f(s) = \max_{\tau \in \Omega} \frac{\sum_{\sigma \in \Omega} v_s(\sigma) K_{f,s}(\sigma, \tau)}{v_s(\tau)}, \quad (65)$$

$$\gamma_s = \max_{\sigma \in \Omega} \frac{b_s(\sigma)}{v_s(\sigma)}. \quad (66)$$

Let D be the graph on F with edges $f \sim g$ and let $\mathcal{M}(D)$ be the trace monoid in which nonadjacent flaws commute. For a trace h , write $N_f(h)$ for the multiplicity of f and $|h| = \sum_f N_f(h)$. Define

$$a_T(s) = \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_{f \in F} \lambda_f(s)^{N_f(h)}, \quad (67)$$

$$P_D(\mathbf{z}) = \sum_{I \in \text{Ind}(F)} (-1)^{|I|} \prod_{f \in I} z_f. \quad (68)$$

The open Shearer region consists of the vectors $\mathbf{z} \geq 0$ for which $P_D(\mathbf{y}) > 0$ for every $0 \leq \mathbf{y} \leq \mathbf{z}$. The Cartier–Foata identity gives

$$\sum_{T \geq 0} a_T(s) z^T = \frac{1}{P_D(z\boldsymbol{\lambda}(s))} \quad (69)$$

as a formal power series, where $\lambda(s) = (\lambda_f(s))_{f \in F}$.

For two formal series $F(z) = \sum_{T \geq 0} a_T z^T$ and $G(z) = \sum_{T \geq 0} b_T z^T$, write $F \preceq_{\text{term}} G$ when $a_T \leq b_T$ for every T .

For positive numbers $(\eta_f)_{f \in F}$, write $\eta(I) = \prod_{f \in I} \eta_f$ and put

$$\theta_s = \max_{f \in F} \frac{\lambda_f(s)}{\eta_f} \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I). \quad (70)$$

Theorem C.1 (Powered matrix-commutative local lemma). *Fix $s > 0$ and suppose that the powered kernels satisfy (63). Then every deterministic nonanticipating selector Λ satisfies*

$$Z_T^\Lambda(s) \leq \gamma_s a_T(s) \quad (T \geq 0), \quad (71)$$

or, equivalently,

$$\sum_{T \geq 0} Z_T^\Lambda(s) z^T \preceq_{\text{term}} \frac{\gamma_s}{P_D(z\lambda(s))}. \quad (72)$$

If $q\lambda(s)$ lies in the open Shearer region for some $q > 1$, then

$$Z_T^\Lambda(s) \leq \frac{\gamma_s}{P_D(q\lambda(s))} q^{-T}. \quad (73)$$

If instead $\theta_s < 1$, then

$$Z_T^\Lambda(s) \leq \gamma_s \left(\sum_{R \in \text{Ind}(F)} \eta(R) \right) \theta_s^T \quad (T \geq 0). \quad (74)$$

All quantities on the right are independent of the selector.

Proof idea. Group execution prefixes by the trace of their repair labels. Lemma C.5 bounds the powered mass of each group by $b_s^\top K_{h,s} \mathbf{1}$, where $K_{h,s}$ is the product of the powered matrices along any word representing the trace h . Commutation makes this product well defined. The positive vector v_s then bounds the group's mass by $\gamma_s \prod_f \lambda_f(s)^{N_f(h)}$. Summing over traces and applying the Cartier–Foata identity gives the term-by-term bound. Appendix C.1 proves the matrix induction, including selectors that remember action labels, and derives the Shearer and cluster estimates.

The term-by-term trace bound is the sharpest of these statements. The Shearer condition turns it into an exponential estimate, while the cluster expression is a convenient local sufficient condition. At $s = 1$ they recover established commutative-LLL bounds. The new use is to control the full family of powered action probabilities, and hence weak randomness and exceptional-tape dimension, by the same matrices.

Whenever either strict condition in Theorem C.1 holds, choose constants such that

$$Z_T^\Lambda(s) \leq C_s \vartheta_s^T, \quad C_s < \infty, \quad 0 < \vartheta_s < 1, \quad (75)$$

using the corresponding constants from (73) or (74). If the rate supplied by the theorem is zero, the same estimate remains valid after replacing it by any fixed number in $(0, 1)$.

Corollary C.2 (Dimension and pointwise coding). *Assume (75) and $\rho_{\max} = \max_{\sigma, f, a} \rho_f(a \mid \sigma) < 1$. Then*

$$\dim_H^{P_{\text{tape}}} \mathcal{N} \leq s. \quad (76)$$

Suppose in addition that s , the process, the selector, and computable numbers $\widehat{C}_s \geq C_s$ and $0 < \widehat{\vartheta}_s < 1$ with $\widehat{\vartheta}_s \geq \vartheta_s$ are computable. Include fixed descriptions of these objects in instance. Every $w \in \mathcal{L}_T^\Lambda$ then satisfies

$$K(w \mid T, \text{instance}) \leq s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1), \quad (77)$$

and every $\omega \in \mathcal{N}$ has $\text{Dim}_{\text{eff}, P}(\omega) \leq s$ in this action-outcome source tree.

The metric in this corollary is tied to complete oracle actions. A binary random-bit implementation gives the same conclusion when it is related to this tree by a computable prefix-free sampler with bounded distortion; no invariance under arbitrary implementations is asserted.

Corollary C.3 (Dependent repair choices). *Let ν be a possibly history-dependent law that uses only actions available at the corresponding reference history. If a constant $D < \infty$ satisfies*

$$\nu[w] \leq D P_{\text{tape}}[w]^s \quad (w \in \mathcal{L}_T^\Lambda, T \geq 0), \quad (78)$$

then

$$\mathbb{P}_\nu\{\tau \geq T\} \leq D C_s \vartheta_s^T. \quad (79)$$

When $0 < s \leq 1$ and the reference oracle is uniform on the same M_t actions available at step t , the explicit condition

$$\max_a \mathbb{P}_\nu\{A_t = a \mid \mathcal{H}_{t-1}\} \leq M_t^{-s}$$

for every feasible positive-probability history implies (78) with $D = 1$ when ν and P_{tape} also use the same initial law. The repair choices may be correlated across time, but may alter neither the selector nor the valid action sets. For a different initial law, or for $s > 1$, the corresponding initial likelihood ratio is absorbed into D when it is bounded.

C.1 Powered matrix theorem

Two familiar conditions imply this matrix identity. First, suppose that for nonadjacent f, g , action-labelled paths

$$\sigma \xrightarrow{f, a} \sigma' \xrightarrow{g, b} \tau$$

are in product-preserving bijection with paths

$$\sigma \xrightarrow{g, b'} \sigma'' \xrightarrow{f, a'} \tau,$$

so that

$$\rho_f(a \mid \sigma) \rho_g(b \mid \sigma') = \rho_g(b' \mid \sigma) \rho_f(a' \mid \sigma''). \quad (80)$$

Second, suppose that each enabled instance of flaw f has the same number M_f of equiprobable actions.

Lemma C.4 (Two routes to powered commutation). *The product-preserving diamond property implies (63) for every $s > 0$. Alternatively, if each enabled f has exactly M_f actions of probability $1/M_f$, then*

$$K_{f,s} = M_f^{1-s} K_{f,1}. \quad (81)$$

Consequently ordinary matrix commutation at $s = 1$ implies (63) for every $s > 0$ in this fixed-arity case. Different action labels may lead to the same terminal state.

We prove the lemma below.

Ordinary matrix commutation at $s = 1$ need not survive powering. For example,

$$A_0 = \begin{pmatrix} 4/5 & 1/5 \\ 2/5 & 3/5 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 7/10 & 3/10 \\ 3/5 & 2/5 \end{pmatrix}$$

commute, whereas their entrywise square roots do not. Thus the full family of powered action probabilities requires additional structure beyond ordinary matrix commutation; Lemma C.4 gives two sufficient conditions.

Proof of Lemma C.4. The (σ, τ) entry of $K_{f,s} K_{g,s}$ is the sum over all action-labelled two-step paths

$$\sigma \xrightarrow{f,a} \sigma' \xrightarrow{g,b} \tau$$

of

$$(\rho_f(a \mid \sigma) \rho_g(b \mid \sigma'))^s.$$

The diamond bijection maps these paths onto the corresponding g, f paths with the same endpoints and preserves each product before it is powered. The two matrix entries are therefore equal.

For the fixed-arity statement, let $m_f(\sigma, \tau)$ be the number of action labels that move σ to τ when f is addressed. If f is present at σ , then

$$K_{f,s}(\sigma, \tau) = m_f(\sigma, \tau) M_f^{-s} = M_f^{1-s} K_{f,1}(\sigma, \tau).$$

Both sides vanish if f is absent. Hence, for nonadjacent f, g whose ordinary matrices commute,

$$K_{f,s} K_{g,s} = M_f^{1-s} M_g^{1-s} K_{f,1} K_{g,1} = K_{g,s} K_{f,s}.$$

□

Write $f \simeq g$ when $f = g$ or $f \sim g$. For a flaw word $\ell = (f_1, \dots, f_T)$, its *full history DAG* has vertex set $[T]$ and an edge $i \rightarrow j$ exactly when $i < j$ and $f_i \simeq f_j$. For such a DAG H , put

$$K_{\emptyset,s} = I, \quad K_{H,s} = K_{L(x),s} K_{H-x,s},$$

where x is any source and $L(x)$ is its label. Under (63), this definition is independent of the chosen source.

Two full history DAGs are identified when they are related by a label-preserving directed-graph isomorphism.

At $s = 1$, Harris, Iliopoulos, and Kolmogorov [HIK25, Lemma 3.5] bound the probability that a prescribed history DAG appears during a run. Their induction allows the current repair to be absent from that DAG and uses substochasticity in this case. Here H records the entire next T -repair prefix, so the current repair must label a source of H . The omission case disappears.

At a fixed power, scalar normalization makes every powered kernel sub-Markov. The existing matrix method can then be applied to the normalized process. Selectors that remember action labels cause no fundamental obstruction: after sampling an endpoint, sample a compatible label with probability proportional to its powered action probability, and retain it in the selector's private history. The direct induction below avoids this auxiliary process. We use it as a transparent route to the prefix power sums, not as a separation from the scope of the matrix-commutative method.

Lemma C.5 (Exact-prefix matrix bound). *Fix $s > 0$ for which (63) holds. For a full history DAG H with T vertices, let $\mathcal{L}_T^\Lambda(H)$ be the prefixes completing T repairs whose labels induce H . Every deterministic nonanticipating selector satisfies*

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq b_s^\top K_{H,s} \mathbf{1}. \quad (82)$$

No injectivity of the action-to-state map is required.

Proof of Lemma C.5. First observe that $K_{H,s}$ is well defined. If x and y are two sources of H , their labels are nonadjacent; otherwise H would contain an edge between them. Their powered matrices commute, and induction on $|H|$ shows that deleting the sources in either order gives the same product.

A stronger conditional statement is proved by induction. Fix an arbitrary past history $\mathfrak{h}$ ending at state σ , and let $M_{\mathfrak{h}}(H)$ be the powered mass of all continuations whose next $|H|$ repairs induce H . The claim is

$$M_{\mathfrak{h}}(H) \leq e_\sigma^\top K_{H,s} \mathbf{1}. \quad (83)$$

It is immediate for $H = \emptyset$.

Suppose $H \neq \emptyset$, and let f be the flaw selected after $\mathfrak{h}$. If no source of H is labelled f , then $M_{\mathfrak{h}}(H) = 0$. Otherwise there is a unique such source x , since two vertices with the same label are adjacent and cannot both be sources. After an action $a \in \mathcal{A}_f(\sigma)$ leading to $\tau = \Phi_f(\sigma, a)$, the remaining repairs must induce $H - x$. Applying the induction hypothesis separately after every possible action gives

$$\begin{aligned} M_{\mathfrak{h}}(H) &\leq \sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma)^s e_{\Phi_f(\sigma, a)}^\top K_{H-x, s} \mathbf{1} \\ &= e_\sigma^\top K_{f, s} K_{H-x, s} \mathbf{1} = e_\sigma^\top K_{H, s} \mathbf{1}. \end{aligned}$$

Actions leading to the same state are added in $K_{f, s}$; no recovery of an action from its endpoints is used. The induction is uniform over $\mathfrak{h}$, so the selector may use the complete past action history.

Finally, sum (83) over the initial source symbols. Their powered masses at state σ sum to $b_s(\sigma)$, which gives (82). $\square$

Proof of Theorem C.1. For every full history DAG H , the definition of $\lambda_f(s)$ gives

$$v_s^\top K_{H, s} \mathbf{1} \leq \left(\prod_{x \in H} \lambda_{L(x)}(s) \right) v_s^\top \mathbf{1} = \prod_{x \in H} \lambda_{L(x)}(s).$$

Since $b_s \leq \gamma_s v_s$ coordinatewise, Lemma C.5 yields

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq \gamma_s \prod_{x \in H} \lambda_{L(x)}(s). \quad (84)$$

The full history DAG of a flaw word is its labelled dependence graph. Full history DAGs, understood up to the label-preserving isomorphism fixed above, are therefore in one-to-one correspondence with traces; only a subset need be realizable by the algorithm. Summing (84) over realized DAGs and then enlarging to all traces of length T proves

$$Z_T^\Lambda(s) \leq \gamma_s \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_f \lambda_f(s)^{N_f(h)} = \gamma_s a_T(s).$$

The Cartier–Foata identity gives the asserted term-by-term formal-series inequality.

If $q\lambda(s)$ lies in the open Shearer region, the series in (69) converges at $z = q$ and has nonnegative terms. Hence

$$q^T a_T(s) \leq \sum_{t \geq 0} q^t a_t(s) = \frac{1}{P_D(q\lambda(s))},$$

which proves (73).

It remains to derive the local cluster estimate. Assume first that $0 < \theta_s < 1$ and set

$$\bar{\lambda}_f = \frac{\lambda_f(s)}{\theta_s}.$$

By definition of θ_s ,

$$\bar{\lambda}_f \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I) \leq \eta_f.$$

The restricted cluster-expansion enumeration therefore gives, for every independent sink set R ,

$$\sum_{\substack{h \in \mathcal{M}(D) \\ \text{sink}(h) = R}} \prod_f \bar{\lambda}_f^{N_f(h)} \leq \eta(R). \quad (85)$$

This is the standard trace-DAG form of the cluster bound; see Harris, Iliopoulos, and Kolmogorov [HIK25, Proposition 3.9]. Restricting the sum to traces of length T and restoring the factor θ_s^T gives

$$a_T(s) \leq \theta_s^T \sum_{R \in \text{Ind}(F)} \eta(R).$$

Together with (71), this proves (74). If $\theta_s = 0$, then every $\lambda_f(s) = 0$, so the assertion is immediate for $T \geq 1$; at $T = 0$ it follows from $\sum_{\sigma} b_s(\sigma) \leq \gamma_s$. $\square$

Proof of Corollary C.2. The depth- T cylinders indexed by $\mathcal{L}_T^\Lambda$ cover $\mathcal{N}$ and have source diameters at most a fixed initial factor times $\rho_{\max}^T$. By (75),

$$\sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s \leq C_s \vartheta_s^T \longrightarrow 0.$$

This proves that the source-relative s -dimensional Hausdorff measure of $\mathcal{N}$ is zero.

For the effective statement, define on $\mathcal{L}_T^\Lambda$

$$\kappa_{T,s}(w) = \frac{P_{\text{tape}}[w]^s}{\widehat{C}_s \widehat{\vartheta}_s^T}.$$

These are subprobability masses because

$$\sum_{w \in \mathcal{L}_T^\Lambda} \kappa_{T,s}(w) \leq \frac{C_s \vartheta_s^T}{\widehat{C}_s \widehat{\vartheta}_s^T} \leq 1.$$

They are uniformly computable under the stated assumptions. Prefix-free coding gives

$$\begin{aligned} K(w \mid T, \text{instance}) &\leq -\log_2 \kappa_{T,s}(w) + O(1) \\ &= s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1). \end{aligned}$$

Removing the condition T costs $O(\log T)$. Since $\rho_{\max} < 1$, the source self-information of a depth- T prefix grows at least linearly in T . Divide by this self-information and take a limsup; the term $T \log_2 \widehat{\vartheta}_s$ is nonpositive, and the result follows. $\square$

Proof of Corollary C.3. The cylinders indexed by $\mathcal{L}_T^\Lambda$ are disjoint. Hence

$$\mathbb{P}_\nu\{\tau \geq T\} = \sum_{w \in \mathcal{L}_T^\Lambda} \nu[w] \leq D \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s.$$

Apply (75). □

D Further applications

For a variable-model event A , let $v(A)$ be its resampled coordinates and let $\pi_{v(A)}$ be their product law. Define its local powered probability mass by

$$r_A(s) = \sum_{b \in A} \pi_{v(A)}(b)^s. \quad (86)$$

Here A is identified with its local configurations on $v(A)$. Taking $v_s(\sigma)$ proportional to $\pi(\sigma)^s$ in (65) gives the exact local charge

$$\lambda_A(s) = r_A(s). \quad (87)$$

Thus Theorem C.1 can be checked from event scopes, local source masses, and the causality graph, without enumerating all assignments. Replacing the independent-set sum in (70) by $\prod_{B \in \Gamma(A)} (1 + \eta_B)$ gives a simpler polynomial-time sufficient test, though generally a weaker one. The independent-set refinement is the standard cluster-expansion form [BFPS11, Peg14]; the use made here of the full family of powered action probabilities is separate.

For $d \geq 2$, define

$$A_d = 1 + \frac{d^d}{(d-1)^{d-1}}. \quad (88)$$

The graph-specific theorem is sharper but involves P_D . The following local relaxation has explicit constants and, relative to the worst-case max-degree bound c_d , loses at most $\log_2(A_d/c_d) \leq \log_2(5/4)$ bits per repair.

Theorem D.1 (An explicit max-degree tail for k -SAT). *Let Φ be a k -CNF on n variables with m clauses, each containing k distinct variables and overlapping at most $d \geq 2$ other clauses. Start from an arbitrary assignment and use any deterministic nonanticipating rule to select a violated clause. Suppose that, at every feasible history, the next k -bit redraw B_t has worst-case conditional min-entropy at least h , in the pointwise sense*

$$\max_b \mathbb{P}\{B_t = b \mid \mathcal{H}_{t-1}\} \leq 2^{-h}. \quad (89)$$

If $\varepsilon = h - \log_2 A_d > 0$, then

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left(\frac{d}{d-1} \right)^m 2^{-\varepsilon T}. \quad (90)$$

For independent uniform redraws, this becomes

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left(\frac{d}{d-1} \right)^m \left(\frac{A_d}{2^k} \right)^T. \quad (91)$$

Writing P for the uniform independent repair-tape law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 A_d}{k} \right\}. \quad (92)$$

If the selector is computable, the same upper bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$. The tail and Hausdorff-dimension conclusions are uniform over every deterministic nonanticipating selector and every history-dependent redraw law satisfying (89).

Corollary D.2 (Fresh fair bits). *Before repair t , let the past determine an injective map*

$$E_t : \{0, 1\}^r \longrightarrow \{0, 1\}^k, \quad 1 \leq r \leq k,$$

then read r fresh independent fair bits U_t and redraw the selected clause to $E_t(U_t)$. If $\varepsilon_r = r - \log_2 A_d > 0$, then (90) holds with $\varepsilon = \varepsilon_r$, and

$$\mathbb{E}\tau \leq \left\lceil \frac{n + m \log_2(d/(d-1))}{\varepsilon_r} \right\rceil + \frac{1}{2^{\varepsilon_r} - 1}. \quad (93)$$

The expected number of fair bits is $r\mathbb{E}\tau$. In particular, if $A_d \leq 2^{k-1}$, taking $r = \lceil \log_2 A_d + 1 \rceil$ gives

$$\mathbb{E}[\text{fair bits}] = O((n + m/d) \log(d+1)). \quad (94)$$

The redraw map may change adaptively with the entire past, but must be fixed before U_t is read.

Harris and Srinivasan [HS19] develop partial resampling by choosing a random subset of a violated event's variables according to a prescribed fractional hitting-set. Their method uses the event structure to improve algorithmic applications. Here the choice of r coordinates may depend arbitrarily on the past, provided it is made before the fresh bits are read and $r > \log_2 A_d$; this is one special case of an injective redraw map.

Messner and Thierauf's *Modified-Search* instead redraws a violated k -clause uniformly from its $2^k - 1$ satisfying local assignments and obtains the corresponding $c_d \leq 2^k - 1$ condition under their lopsided degree convention [MT12]. Corollary D.2 addresses a different resource question: the adaptive redraw may use only 2^r outcomes, and hence only $r = O(\log d)$ fresh fair bits, without requiring that the violated local assignment be excluded. Cs6ka, Grabowski, M6th6, Pikhurko, and Tyros [CGM⁺22] obtain a constant expected total number of random bits for classes with uniformly subexponential graph growth and fixed positive slack in their local-lemma condition. They reuse bits across distant parallel repairs. The present result instead uses fresh conditional min-entropy, applies to arbitrary bounded-degree graphs, and keeps the original sequential repair map; it does not capture that reuse.

Corollary D.3 (A Santha–Vazirani tape). *Let the bits (U_j) satisfy, for some $0 < \alpha \leq 1/2$,*

$$\alpha \leq \mathbb{P}\{U_j = 1 \mid U_1, \dots, U_{j-1}\} \leq 1 - \alpha$$

at every positive-probability history [SV86]. Feed successive blocks of r bits through the adaptive injective maps in Corollary D.2. If

$$\varepsilon_\alpha = r \log_2 \frac{1}{1 - \alpha} - \log_2 A_d > 0,$$

then (90) and (93) hold with ε_α in place of ε_r . Taking $r = k$ and E_t to be the identity runs ordinary clause resampling directly from the Santha–Vazirani tape, without an extractor.

Proposition D.4 (A one-step min-entropy obstruction). *Let $d \geq 2$ and $k \geq 2$. If $d + 1 \leq 2^{k-1}$, there is a satisfiable k -CNF of dependency degree d whose repair scopes overlap without being equal. From a specified initial assignment, uniform independent k -bit redraws satisfy*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = \frac{d+1}{2^k}, \quad (95)$$

$$\dim_H^P \mathcal{N} = \sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega) = \frac{\log_2(d+1)}{k}. \quad (96)$$

There is also a fixed clause-wise redraw source of conditional block min-entropy $\log_2(d+1)$ on which every run is infinite. The dependency graph of this instance is K_{d+1} , for which $\kappa_{K_{d+1}} = d+1$. Thus Theorem 4.1 is sharp for this graph: strict inequality above $\log_2(d+1)$ gives exponential termination, whereas equality can sustain an infinite run. Among guarantees that use only the maximum degree d , the one-step sufficient and obstructing thresholds are separated by less than

$$\log_2 c_d - \log_2(d+1) < \log_2 e$$

bits. When $d+1 = 2^r$, the infinite-run source is an injective r -fair-bit redraw. This last source need not be Santha–Vazirani, so the proposition does not claim sharpness for Corollary D.3.

Corollary D.5 (Rainbow perfect matchings). *Let $n \geq 2$. Color the edges of K_{2n} so that each color occurs on at most $q \geq 2$ edges, and run Kolmogorov’s perfect-matching repair oracle from an arbitrary perfect matching. A flaw is a pair of vertex-disjoint edges of the same color. Put*

$$A = (2n-1)(2n-3), \quad C = (2n-1)(q-1).$$

Suppose, for some $0 \leq h \leq \log_2 A$, every repair choice has conditional atom probability at most 2^{-h} at every feasible positive-probability history. If, for some $\varepsilon > 0$,

$$h \geq \log_2 \left(\frac{256}{27} C \right) + \varepsilon,$$

then every deterministic nonanticipating selector satisfies

$$\mathbb{P}\{\tau \geq T\} \leq (2n-1)!! e^{n/6} 2^{-\varepsilon T}. \quad (97)$$

For uniform repair choices this is nonvacuous when $q-1 < 27(2n-3)/256$. Writing P for the uniform oracle-action law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2((256/27)(2n-1)(q-1))}{\log_2((2n-1)(2n-3))} \right\}. \quad (98)$$

The same bound holds for $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$ under the computability assumptions of Corollary C.2.

The matching oracle is a non-product-space example: its action count, atomicity, and strong commutativity are established by Kolmogorov [Kol18]. The dimensions in (98) refer to its uniform action-outcome tree. The derivations of these consequences are given in Appendix D.3.

D.1 The guarded SAT obstruction

Proof of Theorem 4.2. Let $l(r)$ be the largest integer such that every r -CNF with maximum clause-overlap degree at most $l(r)$ is satisfiable. By Gebauer, Szabó, and Tardos [GST16, Theorem 1.2], there is an unsatisfiable r -CNF F_r of maximum degree $d_r = l(r) + 1$ satisfying (36). Its clauses contain r distinct variables. For each clause C introduce a fresh variable z_C and replace C by $C \vee z_C$. The resulting formula Ψ_r is satisfied when every $z_C = 1$. Its clause-overlap graph is unchanged, since each new variable occurs in only one clause.

Fix an ordering of the r old variables in each clause and place z_C last in its redraw block. Use independent blocks $(U_t, 0)$, where U_t is uniform on $\{0, 1\}^r$. Every private variable remains zero, so any current assignment violates some clause of F_r and hence of Ψ_r . Termination is therefore impossible, regardless of the selector or the realized blocks. Each redraw has exactly 2^r equiprobable outcomes conditionally on the entire past, proving the entropy claim.

For the uniform source, couple the last coordinates of successive redraw blocks to independent Bernoulli(1/2) variables, continued after the run stops. A last coordinate equal to one permanently satisfies the selected clause: its private variable occurs nowhere else, and a satisfied clause cannot be selected for repair. Distinct such successes must therefore belong to distinct clauses. The computation stops no later than the M_r -th success, whose waiting time has mean $2M_r$. This proves (37). For $T \geq 4M_r$, its binomial event is contained in $\{\text{Bin}(T, 1/2) \leq T/4\}$, whose probability is at most $\exp(-T/16)$ by the standard Chernoff bound.

Finally, the symmetric Shearer bound implies $c_{d_r} \geq 2^r$, since otherwise the unsatisfiable formula F_r would be satisfiable. Also,

$$\log_2 c_{d_r} = \log_2 d_r + (d_r - 1) \log_2 \left(1 + \frac{1}{d_r - 1} \right) = r + O(r^{-1/2}).$$

This proves (38). Given any $\eta > 0$, for all sufficiently large r the infinite-run source has entropy $r > \log_2 c_{d_r} - \eta$. Thus replacing the sufficient threshold by $\log_2 c_d - \eta$ would fail on this family. $\square$

D.2 Tree sharpness: full statement

Theorem D.6 (Tree formulas approach the max-degree boundary). *Fix integers $2 \leq d \leq k$. For every height $\ell \geq 1$, there is a satisfiable k -CNF $\Phi_{d,k,\ell}$ whose clause-overlap graph is the full rooted $(d-1)$ -ary tree $T_{d,\ell}$ of height ℓ . Every variable occurs in at most two clauses, every clause overlaps at most d others, and every pair of adjacent bad events is disjoint. Draw the initial assignment uniformly, and then use independent uniform redraws; let P denote this sequential tape law. For any deterministic current-assignment selector,*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = 2^{-k} \kappa_{T_{d,\ell}}, \quad (99)$$

$$\dim_H^P \mathcal{N} = \frac{\log_2 \kappa_{T_{d,\ell}}}{k}. \quad (100)$$

If the selector is computable, the right side of (100) is also the maximal effective strong dimension. Moreover,

$$\kappa_{T_{d,\ell}} \uparrow \frac{d^d}{(d-1)^{d-1}} = c_d \quad (\ell \rightarrow \infty). \quad (101)$$

This tree threshold is the classical regular-tree boundary for the independence polynomial [SS05, HSV18]. Thus the max-degree dimension bound in Theorem 4.1 is best possible in the supremum over finite instances, already for tree dependency graphs.

For each fixed ℓ and each fixed selector in the theorem, put $\delta_\ell = \log_2(\kappa_{T_d, \ell})/k$. There are a probability law ν_{δ_ℓ} supported on $\mathcal{N}$ and a finite D_ℓ such that

$$\nu_{\delta_\ell}[w] \leq D_\ell P[w]^{\delta_\ell} \quad (102)$$

for every finite tape prefix w . Consequently the same law satisfies $\nu_{\delta_\ell}[w] \leq D_\ell P[w]^s$ for every $0 \leq s \leq \delta_\ell$. Hence the global cylinder threshold (33) is sharp, including at the critical power. This statement does not assert a matching one-step conditional min-entropy law.

D.3 Weak-randomness applications

For the variable-model calculation, take

$$v_s(\sigma) = \frac{\pi(\sigma)^s}{\sum_{\xi \in \Omega} \pi(\xi)^s}.$$

For $S = v(A)$ and a fixed terminal assignment τ ,

$$\begin{aligned} \sum_{\sigma} v_s(\sigma) K_{A,s}(\sigma, \tau) &= v_s(\tau) \sum_{b \in A} \pi_S(b)^s \\ &= v_s(\tau) r_A(s). \end{aligned}$$

This proves (87).

Proof of Theorem 4.1. For uniform k -clause resampling, $\lambda_A(s) = 2^{-ks}$ for every clause A . A fixed initial assignment and the uniform choice of v_s give $\gamma_s = 2^n$. Therefore Theorem C.1 gives

$$Z_T^\Lambda(s) \leq 2^n 2^{-ksT} |\{h \in \mathcal{M}(D) : |h| = T\}|.$$

For every $\xi > \kappa_D$, the Cartier–Foata series at $z = \xi^{-1}$ gives

$$|\{h \in \mathcal{M}(D) : |h| = T\}| \leq \frac{\xi^T}{P_D(\xi^{-1} \mathbf{1})}.$$

Multiplying by D_0 proves (32). A conditional atom bound implies $\nu[w] \leq 2^{-hT} = P[w]^{h/k}$ by the chain rule, so this is the special case $s = h/k$ and $D_0 = 1$.

For every $s > \log_2(\kappa_D)/k$, choose $\xi \in (\kappa_D, 2^{ks})$. The resulting prefix power sum decays exponentially, which proves the Hausdorff and, under computability, effective dimension bounds by Corollary C.2. Finally, the symmetric form of Shearer’s theorem [She85, SS05] states that a graph of maximum degree $d \geq 2$ has $z\mathbf{1}$ in its open Shearer region whenever

$$0 \leq z < \frac{(d-1)^{d-1}}{d^d}.$$

Hence $R_D \geq (d-1)^{d-1}/d^d$ and $\kappa_D \leq c_d$. □

Proof of Corollary 4.3. Fix a nonempty $R \subseteq F$. If $\text{Rec}(\omega) = R$, every clause outside R is repaired only finitely often. After a finite tape prefix, the run therefore repairs only clauses in R . Fix such a prefix u . On a history of the instance that retains only the clauses in R , prepend u and follow the original selector until it first asks to repair a clause outside R . At that point, permanently switch to the first currently violated clause of R in a fixed order, stopping if none exists. This defines a deterministic nonanticipating selector for the reduced instance and agrees with the original run

on every continuation after u whose later repair labels all lie in R . It is computable whenever the original selector is. The proof of Theorem 4.1 now applies to the induced graph $D[R]$. For every $s > \log_2(\kappa_{D[R]})/k$, the prefix power sum of the suffix decays exponentially.

There are countably many finite prefixes after which the run remains in R . Prefixing rescales the source metric by a constant, and Hausdorff dimension is stable under countable unions. This proves the setwise bound. In the computable case, a fixed prefix and the finite set R cost only an additive constant in a program description. Prefix-free coding of the same suffix bound and passage to the infimum over s prove (41). $\square$

Proof of Theorem D.1. For a violated k -clause under the uniform reference source, precisely one local assignment is bad, so its powered charge is 2^{-ks} . Take v_s uniform on the 2^n assignments. A fixed initial assignment gives $\gamma_s = 2^n$. Set $\eta_A = \eta = 1/(d-1)$. A clause is adjacent to every other member of its inclusive neighborhood. An independent subset of that neighborhood therefore either consists of the clause itself or is contained in its at most d neighbors. Hence

$$\sum_{I \in \text{Ind}(\Gamma(A))} \eta(I) \leq \eta + (1 + \eta)^d.$$

Consequently

$$\theta_s \leq 2^{-ks} \frac{\eta + (1 + \eta)^d}{\eta} = 2^{-ks} \left(1 + \frac{d^d}{(d-1)^{d-1}} \right) = 2^{-ks} A_d,$$

whereas

$$\sum_{R \in \text{Ind}(F)} \eta(R) \leq (1 + \eta)^m = \left(\frac{d}{d-1} \right)^m.$$

The atom condition implies that every length- T redraw word has probability at most 2^{-hT} . With $s = h/k$, this equals its uniform reference probability raised to s . If $h = \log_2 A_d + \varepsilon$, apply Corollary C.3 to obtain (90).

Taking $h = k$ gives (91). For the dimension statement, every $s > \log_2 A_d/k$ makes $\theta_s < 1$. Corollary C.2 applies for every such s ; let s decrease to the displayed value. The bound by one is automatic. $\square$

Proof of Corollary D.2. Conditioned on the past, injectivity of E_t makes every supported redraw have probability exactly 2^{-r} . The encoder is chosen before U_t , so its dependence on the past does not change this fact. Apply Theorem D.1 with $h = r$. For a bound $\mathbb{P}\{\tau \geq T\} \leq C_0 2^{-\varepsilon T}$, bound the first $\lceil \log_2(C_0)/\varepsilon \rceil$ terms of $\mathbb{E}\tau = \sum_{T \geq 1} \mathbb{P}\{\tau \geq T\}$ by one and sum the remaining geometric series. With $C_0 = 2^n(d/(d-1))^m$ this gives (93); every repair consumes exactly r bits.

If $A_d \leq 2^{k-1}$ and $r = \lceil \log_2 A_d + 1 \rceil$, then $r \leq k$, $\varepsilon_r \geq 1$, and $r = O(\log(d+1))$. Moreover, $\log_2(d/(d-1)) = O(1/d)$ for $d \geq 2$. Substitution in (93) proves (94). $\square$

Proof of Corollary D.3. Condition on the entire past before a block is read. The chain rule and the Santha–Vazirani inequalities give, for every $z \in \{0, 1\}^r$,

$$\mathbb{P}\{(U_{j+1}, \dots, U_{j+r}) = z \mid U_1, \dots, U_j\} \leq (1 - \alpha)^r.$$

After an injective map, the same upper bound holds for every redraw atom. Thus Theorem D.1 applies with $h = r \log_2(1/(1 - \alpha))$. $\square$

Proof of Proposition D.4. Put $\ell = \lceil \log_2(d+1) \rceil$ and choose distinct words $u_1, \dots, u_{d+1} \in \{0, 1\}^\ell$. Use ℓ common variables and, for each i , a disjoint private block P_i of $k - \ell$ variables. Let C_i be the unique clause on the common variables and P_i that is false when the common word is u_i and P_i is all zero. The clauses share the common variables, so their dependency graph is K_{d+1} , but their scopes are not equal. Distinct clauses cannot be false simultaneously. The formula is satisfiable, for example by setting one private variable in each P_i to one.

Start with every private block zero and the common word equal to u_1 . Call i active when P_i is still zero and let q be the number of active indices. In every live state the common word is u_i for a unique active i , so C_i is the only violated clause. Write $M = 2^{k-\ell}$. When C_i is repaired, exactly q of its 2^k redraws leave q unchanged: the new private block is zero and the new common word is u_j for an active j . Exactly $(q-1)(M-1)$ redraws move to $q-1$: the new private block is nonzero and the new common word is u_j for one of the other active indices. Every other redraw terminates. Thus the all-power live matrix on $q = 1, \dots, d+1$ is bidiagonal, with

$$R_s(q, q) = q 2^{-ks}, \quad R_s(q, q-1) = (q-1)(M-1)2^{-ks}. \quad (103)$$

Its spectral radius is $(d+1)2^{-ks}$. Proposition E.1 therefore gives (95) and (96), including pointwise attainment.

At $q = d+1$, instead redraw the selected scope uniformly from the $d+1$ assignments $(u_j, 0)$, $1 \leq j \leq d+1$. This source has conditional block min-entropy $\log_2(d+1)$ and remains at $q = d+1$ forever. If $d+1 = 2^r$, the redraw is the image of an injective map applied to r fresh fair bits. Finally,

$$\frac{c_d}{d+1} = \frac{d}{d+1} \left(1 + \frac{1}{d-1}\right)^{d-1} < e,$$

which proves the stated gap. $\square$

Proof of Theorem D.6. Orient every edge of $T_{d,\ell}$ away from the root and introduce one Boolean variable X_e for each edge. For a vertex v , form a clause C_v containing one literal for each incident edge, using opposite signs at the two endpoints. Add $k - \deg(v)$ private variables to C_v , each used in no other clause. This is possible because $\deg(v) \leq d \leq k$. Every clause has exactly k distinct variables, and every variable occurs once or twice.

Two clauses share a variable exactly when their vertices are adjacent. Their bad events prescribe opposite values to the shared edge variable and are therefore disjoint; nonadjacent clauses have disjoint scopes. The formula is satisfiable. For example, use the sign convention in which the parent literal on an edge is X_e and the child literal is $\neg X_e$, set every $X_e = 0$, and set one private literal of the root to true. Every nonroot clause is satisfied by its parent-edge literal, while the root has a private literal because its degree is $d-1 < k$.

Every bad event is atomic and has probability 2^{-k} . Let

$$N_T(D) = |\{h \in \mathcal{M}(D) : |h| = T\}|.$$

Let n be the number of variables and fix the selector. Every depth- T prefix has probability $2^{-(n+kT)}$. Fix a trace H and a satisfying terminal assignment y . Reconstruct the prefix by induction on $|H|$. For $H = \emptyset$, the reconstructed state is y . Otherwise choose a source x of H , first reconstruct $H - x$, and then restore on the scope of $L(x)$ the unique local assignment that makes $L(x)$ bad; atomicity gives uniqueness. The redraw block associated with x is the restriction of the reconstructed post-repair assignment to the scope of $L(x)$; hence both the predecessor state and the corresponding tape symbol are determined. The result does not depend on the chosen source. Any two source-removal orders are connected by swaps of adjacent incomparable nodes, and incomparable nodes have

nonadjacent labels and hence disjoint scopes. Their reverse restorations, including the recorded redraw blocks, therefore commute.

Inductively, the bad events in the reconstructed initial state are exactly the labels of the sources of H . The restored event $L(x)$ is bad; every adjacent bad event is absent by extremality, while every nonadjacent event is unchanged and is governed by the induction hypothesis for $H - x$. Nodes with the same label are comparable, so each source label occurs only once. The selector therefore chooses the label of a unique source node. Removing that node and repeating produces a unique selector-compatible linear extension of H , and hence exactly one live prefix ending at y . Different traces give different repair traces. Conversely, a trace and a terminal assignment determine at most one live prefix, and there are at most 2^n terminal assignments. This is the elementary finite-horizon form of the extremal probability-weighted trace identity [GJL19, Jer24]. Hence, for every $s > 0$,

$$2^{-sn}2^{-ksT}N_T(T_{d,\ell}) \leq Z_T^\Lambda(s) \leq 2^{(1-s)n}2^{-ksT}N_T(T_{d,\ell}). \quad (104)$$

The limit of the trace growth exists without an aperiodicity assumption. Choose one canonical word for each trace. Splitting the canonical word of a length- $(m+n)$ trace after m letters maps it injectively to a pair of traces of lengths m and n : the pair determines its product and hence the original trace. Therefore

$$N_{m+n}(D) \leq N_m(D)N_n(D).$$

Fekete's lemma applied to $\log N_T(D)$ shows that $N_T(D)^{1/T}$ has a limit, whose value is κ_D by (30). It follows from (104) that

$$\lim_{T \rightarrow \infty} Z_T^\Lambda(s)^{1/T} = 2^{-ks} \kappa_{T_{d,\ell}}.$$

Let $\tilde{Z}_j(s)$ count prefixes still live after j repairs, including the initial assignment. The final redraw in a completed- T prefix is unrestricted, so

$$Z_T(s) = 2^{k(1-s)} \tilde{Z}_{T-1}(s) \quad (T \geq 1).$$

Thus these two sums have the same exponential rate. At $s = 1$ this proves (99). The unique power at which the rate equals one is $\log_2(\kappa_{T_{d,\ell}})/k$. Applying Proposition E.1 to the reachable live components gives (100) and, for a computable selector, effective attainment.

It remains to calculate the trace growth. Put $b = d - 1$ and define

$$r_0(z) = z, \quad r_{j+1}(z) = \frac{z}{(1 - r_j(z))^b}.$$

As long as the denominators are positive, $r_j(z)$ is the signed root-occupation ratio for a full b -ary subtree of height j . The first positive zero $z_{d,\ell}$ of $P_{T_{d,\ell}}(z\mathbf{1})$ is characterized by $r_\ell(z_{d,\ell}) = 1$. The critical value for the iteration is

$$z_* = \max_{0 \leq r < 1} r(1-r)^b = \frac{b^b}{(b+1)^{b+1}} = \frac{(d-1)^{d-1}}{d^d}.$$

If $z < z_*$, the iteration increases to the smaller fixed point of $r = z/(1-r)^b$ and remains below one. If $z > z_*$, there is no fixed point in $[0, 1)$, so the iteration reaches or exceeds one after finitely many steps. Hence $z_{d,\ell} \downarrow z_*$ and

$$\kappa_{T_{d,\ell}} = z_{d,\ell}^{-1} \uparrow z_*^{-1} = c_d.$$

Finally put $\delta_\ell = \log_2(\kappa_{T_{d,\ell}})/k$. For the fixed selector, consider the finite live chain on full assignments. The exact growth calculation above and the finite-state decomposition give a reachable

irreducible component C whose powered transition matrix satisfies $\rho(B_{\delta_\ell, C}) = 1$. Choose a positive right Perron vector h and a live prefix u entering C . On an internal edge $e : x \rightarrow y$ of C , set

$$Q(e \mid x) = P(e)^{\delta_\ell} \frac{h(y)}{h(x)}.$$

Its row sums are one, so prefixing Q by u defines a probability law ν_{δ_ℓ} supported on $\mathcal{N}$. Telescoping along extensions of u , and absorbing the source probability of u and the finitely many ratios of entries of h into D_ℓ , gives (102) for every finite prefix w . Since $P[w] \leq 1$, the same inequality holds with $P[w]^s$ for every $0 \leq s \leq \delta_\ell$. $\square$

Proof of Corollary D.5. For the matching oracle of Kolmogorov [Kol18], every flaw has

$$A = (2n - 1)(2n - 3)$$

equiprobable actions, distinct oracle executions have distinct outputs in the action-expanded graph, and the oracle is strongly commutative. Thus its powered charge is A^{-s} . Put $C = (2n - 1)(q - 1)$. The neighborhood of a flaw is covered by four families, one for each endpoint, each containing at most C flaws. Therefore, for constant η ,

$$\sum_{I \in \text{Ind}(\Gamma(f))} \eta^{|I|} \leq (1 + C\eta)^4.$$

Taking $\eta = 1/(3C)$ gives

$$\theta_s \leq \frac{A^{-s}}{\eta} (1 + C\eta)^4 = \frac{256}{27} C A^{-s}.$$

There are at most

$$|F| \leq \frac{n(2n - 1)(q - 1)}{2}$$

flaws, and hence

$$\sum_{R \in \text{Ind}(F)} \eta^{|R|} \leq (1 + \eta)^{|F|} \leq e^{|F|\eta} \leq e^{n/6}.$$

With v_s uniform on the $(2n - 1)!!$ perfect matchings, a fixed initial matching has $\gamma_s = (2n - 1)!!$. The conditional atom bound and the uniform reference oracle give power domination with $s = h / \log_2 A$. Corollary C.3 now proves (97). Letting s decrease to $\log_2((256/27)C) / \log_2 A$ proves the dimension bound. Under uniform actions, the strict condition $\theta_1 < 1$ is equivalent to $q - 1 < 27(2n - 3)/256$. $\square$

E Finite-state formulas and dimension

For clarity, recall the distinction between Hausdorff and box-counting dimension. In a metric space, set

$$\mathcal{H}_\sigma^a(E) = \inf \left\{ \sum_i (\text{diam } U_i)^a : E \subseteq \bigcup_i U_i, \text{diam } U_i \leq \sigma \right\}.$$

The exponent a is a trial value. Hausdorff dimension is the critical exponent of $\lim_{\sigma \downarrow 0} \mathcal{H}_\sigma^a(E)$; the covering sets may have different sizes. By contrast, if $N(E, \sigma)$ is the smallest number of sets of diameter at most σ needed to cover E , then the lower and upper limits of $\log N(E, \sigma) / \log(1/\sigma)$ define box dimensions. They need not equal Hausdorff dimension [Fal14]. Neither definition requires a finite-dimensional parameterization of E .

For a measure μ , its local dimension at ω , when the limit exists, is $\lim_{r \downarrow 0} \log \mu(B(\omega, r)) / \log r$. This measure-dependent quantity is different from the effective dimension in (39). The finite-state measures below connect them through explicit cylinder bounds.

Let $\mathcal{G}$ be a finite directed multigraph of nonterminal states with fixed initial state s_0 . Each outgoing edge $e : s \rightarrow t$ represents a distinct next source outcome and has probability $p(e) > 0$. The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path $w = e_1 \cdots e_n$, traversing states $s_0, s_1, \dots, s_n$, determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^n p(e_i).$$

Let $\mathcal{W}_n$ be the set of surviving paths of length n from s_0 . Assume that the reachable nonterminal graph is strongly connected, contains a directed cycle, and satisfies $\max_e p(e) < 1$. For $\theta \geq 0$, define

$$(B_\theta)_{st} = \sum_{e:s \rightarrow t} p(e)^\theta, \quad \Lambda(\theta) = \log_2 \rho(B_\theta). \quad (105)$$

For any square matrix C , its spectral radius is $\rho(C) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } C\}$. At $\theta = 1$, B_1 is the substochastic transition matrix among nonterminal states: each row sum is at most one, and the missing mass is the one-step termination probability. The quantity $\Lambda(\theta)$ is the base-two pressure, the exponential growth rate of the total θ -weight of surviving paths. Graphs that are not strongly connected are handled by applying the result to each reachable strongly connected component containing a directed cycle and taking the largest dimension; if there is no such component, the nonterminating set is empty.

For distinct infinite paths put $d_P(\omega, \eta) = P[\omega \wedge \eta]$, where $\omega \wedge \eta$ is their longest common prefix, and set $d_P(\omega, \omega) = 0$. A surviving-path cylinder has diameter at most $P[w]$; it can have smaller diameter when later continuation is forced. The corresponding dimension is the source-relative dimension $\dim_H^P$; for a uniform q -ary source it is Hausdorff dimension in the q -adic metric. For an infinite path $\omega = e_1 e_2 \cdots$, use the corresponding pointwise definition

$$\text{Dim}_{\text{eff}, P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(e_1 \cdots e_n)}{-\log_2 P[e_1 \cdots e_n]},$$

and define $\dim_{\text{eff}, P}$ by replacing the limsup with a liminf.

Proposition E.1 (Detailed finite-state formula). *There is a unique $\delta \in [0, 1]$ satisfying $\rho(B_\delta) = 1$. Moreover,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \sum_{w \in \mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \quad (106)$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \quad (107)$$

$$\dim_H^P \mathcal{N} = \delta. \quad (108)$$

If $B_\delta h = h$ with $h_s > 0$ for every state s , then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \quad (109)$$

is a probability law on surviving paths. For every $w = e_1 \cdots e_n \in \mathcal{W}_n$, define its probability under this law by

$$Q_\delta[w] = \prod_{i=1}^n Q_\delta(e_i \mid s_{i-1}).$$

Then it satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \quad (110)$$

For a computable system with computable Q_δ , every nonterminating tape satisfies $\text{Dim}_{\text{eff},P}(\omega) \leq \delta$. A tape that is Martin–Löf random under Q_δ satisfies $\text{dim}_{\text{eff},P}(\omega) = \text{Dim}_{\text{eff},P}(\omega) = \delta$.

The requirement that Q_δ be computable is the effectivity condition needed for the pointwise conclusion. In the strict-interior finite cases below it follows automatically: if the edge probabilities are computable and

$$\rho(B_0) > 1 > \rho(B_1),$$

then $\theta \mapsto \rho(B_\theta)$ is uniformly computable and strictly decreasing, so bisection computes δ . Irreducibility makes the Perron eigenvalue simple; solving $B_\delta h = h$ with $\sum_s h_s = 1$ gives a computable positive eigenvector and hence a computable Q_δ . Endpoint roots used below are explicit.

Proof of Proposition E.1. Since the graph contains a directed cycle, $\rho(B_0) \geq 1$. Since B_1 is substochastic, $\rho(B_1) \leq 1$. If $\theta' > \theta$, then every nonzero entry contributed by an edge is smaller in $B_{\theta'}$ than in B_θ , because $p(e) < 1$. Strict Perron comparison on the common strongly connected support therefore gives $\rho(B_{\theta'}) < \rho(B_\theta)$. Together with continuity, this proves existence and uniqueness of δ . The relevant matrix identity is

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^\top B_\theta^n \mathbf{1},$$

where e_{s_0} is the coordinate vector of the initial state and $\mathbf{1}$ is the all-ones column vector. Perron–Frobenius theory proves (106); setting $\theta = 1$ gives the same escape exponent for remaining live after n repairs. Completing n repairs requires remaining live after $n - 1$; this shift leaves the exponent in (107) unchanged.

If $\theta > \delta$, then $\rho(B_\theta) < 1$, so the total θ -weight of the length- n surviving cylinders tends to zero. These cylinders cover $\mathcal{N}$, giving $\text{dim}_H^P \mathcal{N} \leq \delta$. At $\theta = \delta$, (109) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because h is positive on a finite state space, there is a constant $C < \infty$, independent of w , such that $Q_\delta[w] \leq CP[w]^\delta$. The mass-distribution principle states that a probability measure satisfying this cylinder bound forces $\text{dim}_H^P \mathcal{N} \geq \delta$. It therefore gives the required lower bound, proving (108) and (110) [Bow75, MW88].

For a computable system with computable Q_δ , coding under Q_δ gives, for every surviving prefix w of length n ,

$$K(w) \leq -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a Q_δ -Martin–Löf-random path [Lut03, AHLM07, Sim15]. $\square$

Under a uniform q -ary source, let $M = (M_{st})$, where M_{st} counts the source symbols that move s to t . Then $B_\theta = q^{-\theta} M$. If

$$\alpha = -\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\}$$

is the survival exponent, then

$$\text{dim}_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Writing $h_{\text{surv}} = \log_2 \rho(M)$ for the topological entropy, in bits per step, of the surviving path space gives

$$\alpha = \log_2 q - h_{\text{surv}}, \quad \dim_H \mathcal{N} = \frac{h_{\text{surv}}}{\log_2 q}.$$

Thus α is the entropy deficit imposed by survival, not an entropy function. Equation (110) is the finite-state form of the comparison used in Theorem F.1.

F A backward identity for individual runs

The source-power theorem sums over all live prefixes. A complementary identity controls each realized run without a commutativity or local-lemma assumption.

Let $X_1, \dots, X_n$ be independent finite-valued variables with product law $\pi = \bigotimes_i \pi_i$. Each bad event A depends on coordinates $v(A)$ and has probability $p_A > 0$. Draw $S_0 \sim \pi$. Whenever a bad event is present, a deterministic rule selects A_t , an independent block $Z_t \sim \pi_{v(A_t)}$ is written on its coordinates, and the resulting assignment is S_t . On the event $E_T = \{\tau \geq T\}$, put

$$\Xi_T = (S_0, Z_1, \dots, Z_T), \quad P_{\text{tape}}(\Xi_T) = \pi(S_0) \prod_{t=1}^T \pi_{v(A_t)}(Z_t).$$

Appendix F.1 gives the formal prefix tree and its source-relative metric.

Let $Y_t = S_{t-1}|_{v(A_t)}$ be the overwritten configuration and let $L_T = (A_1, \dots, A_T)$. Because A_t is true immediately before repair,

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^T p_{A_t} \pi_{v(A_t)}(Y_t \mid A_t). \quad (111)$$

For any subprobability law Q_T that is positive on feasible length- T repair histories, define the backward law

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^T \pi_{v(A_t)}(y_t \mid A_t). \quad (112)$$

The map

$$\Xi_T \mapsto (S_T, L_T, Y_{1:T})$$

is injective: reverse the repairs, reading each used block from the current assignment and then restoring the overwritten values.

Theorem F.1 (Backward likelihood identity). *Every execution in E_T satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \quad (113)$$

Call the right side Δ_T on E_T and set $\Delta_T = -\infty$ otherwise. Then, for every $u \geq 0$,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \quad (114)$$

If the product laws, bad-event predicates and their scopes, and the family $(Q_T)_{T \geq 1}$ are uniformly computable, with fixed descriptions included in instance, then

$$-\log_2 P_{\text{tape}}(\Xi_T) - K(\Xi_T \mid T, \text{instance}) \geq \Delta_T - O(1). \quad (115)$$

The likelihood ratio also gives an exact slack identity. Let $\mathcal{W}_T$ be the prefix-free set of minimal prefixes producing T repairs, let $p_T = P_{\text{tape}}(E_T)$, and put

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)), \quad \mu_T(w) = G_T(R_T(w)), \quad m_T = \sum_{w \in \mathcal{W}_T} \mu_T(w).$$

When $p_T m_T > 0$, define $P_T(w) = P_{\text{tape}}(w)/p_T$ and $\nu_T(w) = \mu_T(w)/m_T$.

Corollary F.2 (Exact slack and the optimal history law). *For every fixed Q_T ,*

$$-\log_2 p_T = \mathbb{E}_{P_T} \Delta_T + D_2(P_T \| \nu_T) - \log_2 m_T. \quad (116)$$

If $J_T = \sum_{t=1}^T \log_2(1/p_{A_t})$, then

$$\sup_{Q_T} \mathbb{E}[\Delta_T \mid E_T] = \mathbb{E}[J_T \mid E_T] - H_2(L_T \mid E_T), \quad (117)$$

where the supremum is over subprobability laws positive on feasible histories. It is attained by $Q_T(\ell) = \mathbb{P}\{L_T = \ell \mid E_T\}$ on the conditional support.

Reverse reconstruction and its Kolmogorov-complexity use are established in entropy compression [Mos09, MT12]. The point of Theorem F.1 is the common nonuniform likelihood: one ratio gives the tail, the tape-by-tape coding saving, and the exact relative-entropy slack. Proofs are in Appendix F.

F.1 The consumed-tape model

The tape is the rooted tree of source choices actually queried by the computation. A finite node is a feasible transcript $w = (s_0, z_1, \dots, z_t)$. The root branches to s_0 with probability $\pi(s_0)$. If w is live and the rule next selects A_{t+1} , its children are indexed by local blocks z , with edge probability $\pi_{v(A_{t+1})}(z)$. Hence

$$P_{\text{tape}}[w] = \pi(s_0) \prod_{j=1}^t \pi_{v(A_j)}(z_j). \quad (118)$$

Terminal transcripts are leaves, and infinite branches are exactly the nonterminating consumed tapes. For two distinct infinite branches, let $\omega \wedge \eta$ be their longest common consumed prefix and set

$$d_P(\omega, \eta) = P_{\text{tape}}[\omega \wedge \eta]. \quad (119)$$

All dimension statements assume that these cylinder masses tend to zero along every infinite branch.

Fixed computable enumerations of the outcome alphabets give a canonical self-delimiting encoding of each transcript. The dimensions are unchanged under a computable prefix-tree isomorphism Ψ with computable inverse and bounded source distortion

$$C^{-1} P_{\text{tape}}[w] \leq P'_{\text{tape}}[\Psi(w)] \leq C P_{\text{tape}}[w],$$

because self-information and prefix complexity then change by at most a bounded additive term. No invariance under arbitrary recodings is asserted. A pre-sampled resampling table is only a coupling device: unused entries are excluded, since adjoining independent unused coordinates can change tape dimension without changing the computation.

F.2 Backward likelihood

Proof of Theorem F.1. Take the ratio of (112) and (111). The final-state factor cancels, as do all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^T p_{A_t}} = 2^{\Delta_T},$$

which proves (113).

It remains to use this likelihood ratio. Let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes that produce at least T repairs, and let $\mathcal{R}_T$ be the set of formal records on which G_T is defined. For $w \in \mathcal{W}_T$, write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)).$$

The reconstruction argument shows that R_T is injective.

First observe that G_T has total mass at most one. Indeed, for a fixed list $\ell = (A_1, \dots, A_T)$, the final-state law and each conditional overwritten-value law have total mass one. Consequently,

$$\sum_{r \in \mathcal{R}_T} G_T(r) = \sum_{\ell} Q_T(\ell) \leq 1.$$

Define the corresponding mass on live tape prefixes by

$$\mu_T(w) = G_T(R_T(w)), \quad w \in \mathcal{W}_T.$$

Injectivity of R_T gives

$$\sum_{w \in \mathcal{W}_T} \mu_T(w) = \sum_{r \in R_T(\mathcal{W}_T)} G_T(r) \leq \sum_{r \in \mathcal{R}_T} G_T(r) \leq 1.$$

Thus μ_T is a subprobability mass function. Its total mass can be strictly less than one because Q_T may itself have mass less than one, or because G_T may assign mass to formal records that are not produced by the algorithm.

For every $w \in \mathcal{W}_T$, the identity already proved gives

$$2^{\Delta_T(w)} = \frac{G_T(R_T(w))}{P_{\text{tape}}(w)} = \frac{\mu_T(w)}{P_{\text{tape}}(w)}.$$

Because $\Delta_T = -\infty$ outside E_T and the cylinders determined by $\mathcal{W}_T$ are disjoint, for every $u \geq 0$,

$$\begin{aligned} \mathbb{P}\{\Delta_T \geq u\} &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} P_{\text{tape}}(w) \\ &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} 2^{-\Delta_T(w)} \mu_T(w) \\ &\leq 2^{-u} \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} \mu_T(w) \\ &\leq 2^{-u}. \end{aligned}$$

This proves (114); equivalently, it is Markov's inequality applied to a subprobability likelihood ratio.

Regard G_T as a subprobability mass on all formal records $(s, \ell, y_{1:T})$, without restricting to records consistent with the selector. Under the stated hypotheses these masses are uniformly computable. Prefix-free coding [Sha48, Fan49, LV19] gives

$$K(R_T(\Xi_T) \mid T, \text{instance}) \leq -\log_2 G_T(R_T(\Xi_T)) + O(1).$$

A uniform computable decoder reverses the repairs: start from s ; for $t = T, \dots, 1$, read z_t from the current assignment on $v(A_t)$, then restore y_t there. The final assignment is s_0 . On a record arising from an execution, this returns exactly $\Xi_T = (s_0, z_1, \dots, z_T)$ without evaluating the selector. Consequently,

$$K(\Xi_T \mid T, \text{instance}) \leq -\log_2 G_T(R_T(\Xi_T)) + O(1) = -\log_2 P_{\text{tape}}(\Xi_T) - \Delta_T + O(1).$$

The constants are uniform in T and the realized run. This proves (115) without requiring a computable selector. $\square$

Proof of Corollary F.2. On $\mathcal{W}_T$,

$$\Delta_T(w) = \log_2 \frac{m_T \nu_T(w)}{p_T P_T(w)}.$$

Taking P_T -expectations gives (116). For the second claim, $\Delta_T = J_T + \log_2 Q_T(L_T)$. If r_T is the conditional law of L_T given E_T , then

$$\sum_{\ell} r_T(\ell) \log_2 Q_T(\ell) = -H_2(r_T) - D_2(r_T \parallel Q_T)$$

when Q_T has total mass one; allocating less mass can only decrease the left side. The maximum is attained at $Q_T = r_T$. $\square$

AI methodologies

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